Learning a Codebase

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April 22, 2026

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- 2.2 Time series sustained components touched by developer - 14 For developer day and components touched. . . . 66

3 Repository: 21 **68**
- 3.1 Time series sustained components touched by developer - 21 For developer commits and components touched. . . . 68
- 3.2 Time series sustained components touched by developer - 21 For developer day and components touched. . . . 71

1 Repository

1.1 Glossary - Summary

- The founder developer starts in the first six months of a project.
- Late joiner developers begin after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.2 Glossary - Summary

- The founder developer starts in the first six months of a project.
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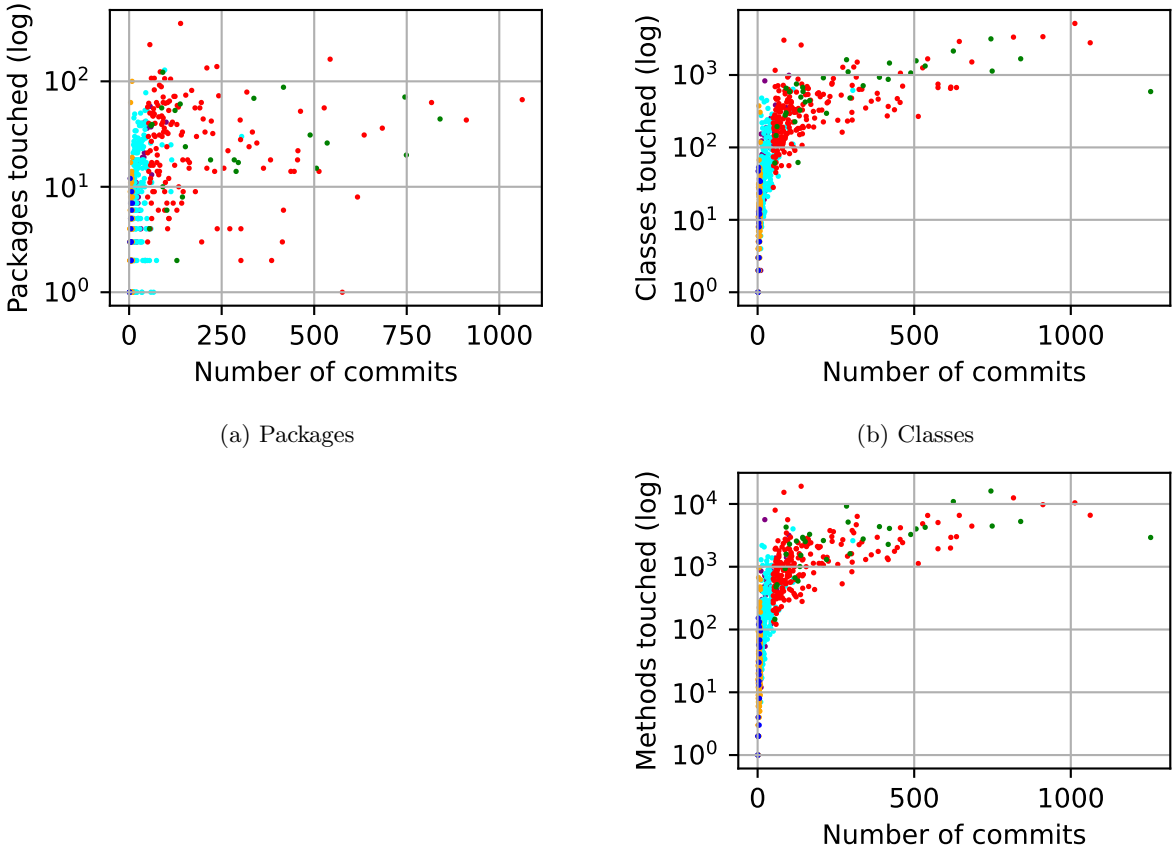
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.3 Repository Summary Table

Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and at least three sustained late joiner developers. Sustained later joiner developers joining the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Joiner	Moderate Joiner	Sustained Joiner	Commit	Start	End
3	activiti	4	7	6	47	20	5	8662	2010-Jun-18	On going
5	airbyte-platform	3	2	1	45	32	6	6784	2020-Aug-04	2023-Sep-27
7	ambari	0	3	0	34	32	15	4247	2011-Sep-22	On going
10	automq	2	0	0	40	10	3	1874	2011-Aug-10	2017-Apr-21
14	buck	8	5	6	119	85	21	14876	2013-Mar-21	2022-May-05
15	camel	0	4	1	257	63	13	36528	2007-Mar-19	On going
18	checkstyle	0	0	1	57	42	9	17338	2001-Jun-22	On going
20	cxf	6	6	3	27	17	3	7893	2008-Apr-29	On going
21	intellij-community	0	3	7	113	93	84	101247	2004-Nov-11	2018-Apr-18
26	guava	0	1	0	47	8	3	9685	2009-Sep-15	On going
28	jenkins	1	1	1	144	39	4	30252	2006-Nov-05	On going
33	openmrs-core	0	1	0	123	26	3	15757	2006-May-03	On going
36	presto	0	1	3	162	71	18	30605	2012-Aug-09	On going
37	quarkus	1	3	5	126	25	8	81612	2018-Jun-14	On going
39	selenium	3	3	0	55	17	8	23354	2004-Nov-03	On going
	Total	28	40	34	1396	580	203	390714		

1.4 Sample scatter touched against commit -



(a) Packages

(b) Classes

(c) Methods

Figure 1: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 from six (6) categories: Transient Founder (Blue, 28), Moderate Founder (Purple, 40), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

1.5 Total commits by developer - Developer Commits

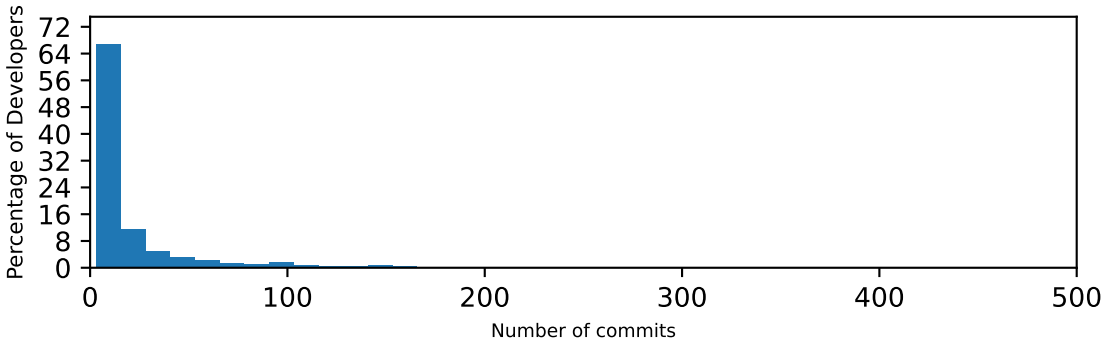
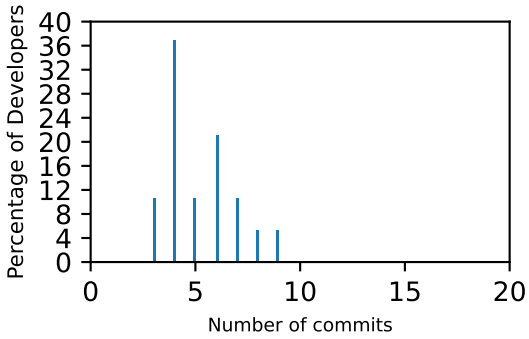
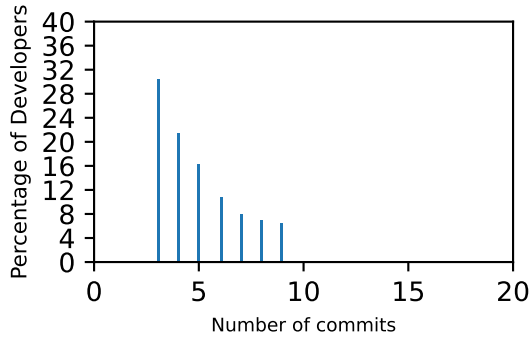


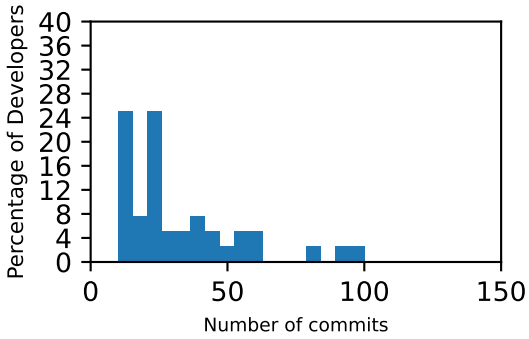
Figure 2: Histogram of the number from all developers (n=1906) (y-axis) against the total number of commits (x-axis) in 15 repositories sampled from GitHub.



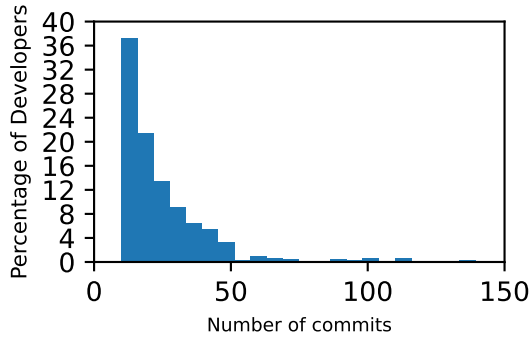
(a) Transient founder (n=19).



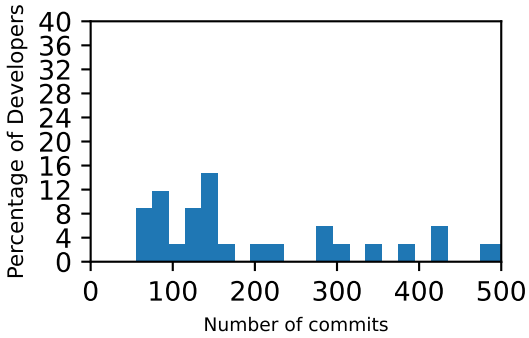
(b) Transient later joiner (n=1030).



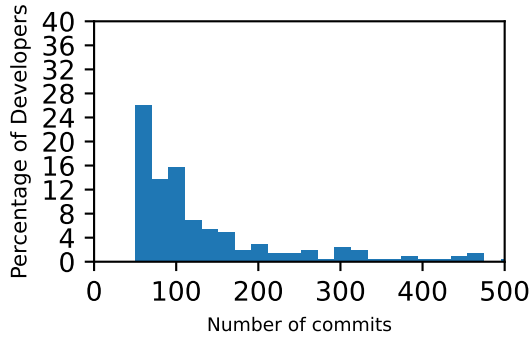
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



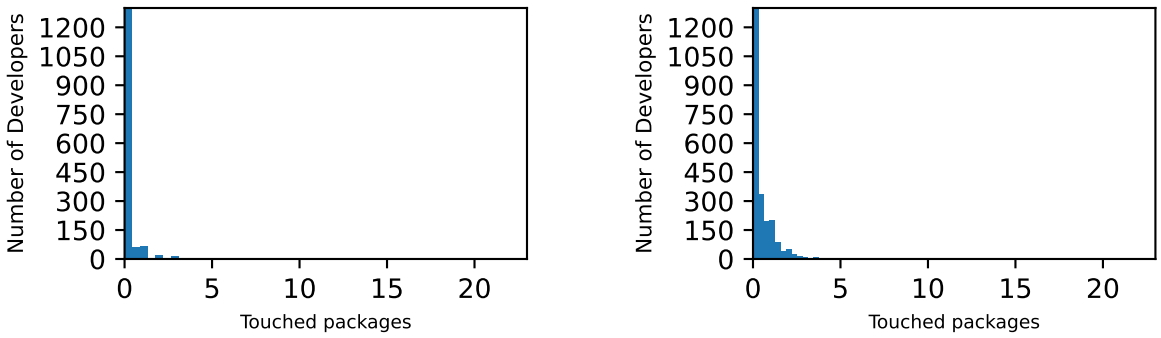
(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 3: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 15 repositories sampled from GitHub.

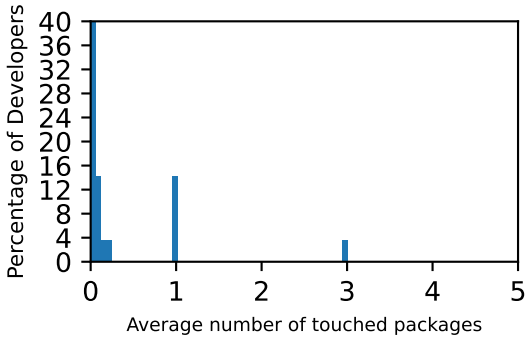
1.6 Histogram components touched by developer - For components with total



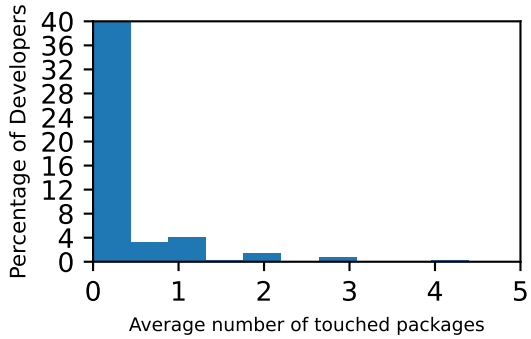
(a) By Day.

(b) By Commit.

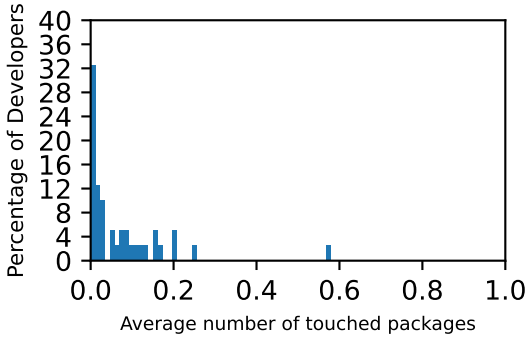
Figure 4: Histogram of the number of developers (y-axis) against the average packages touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



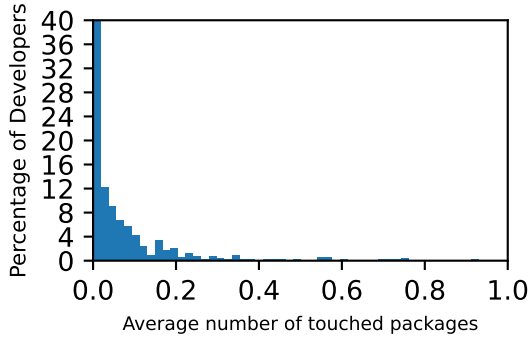
(a) Transient founder (n=28).



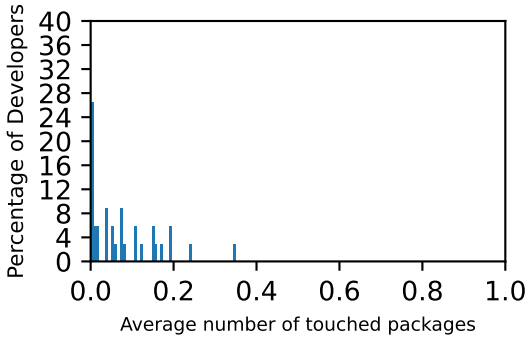
(b) Transient later joiner (n=1396).



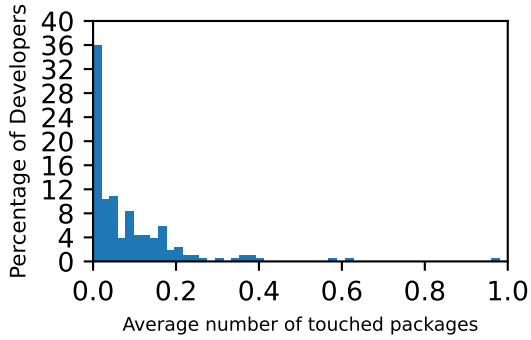
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

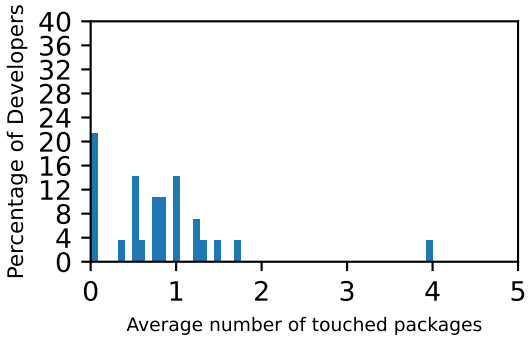


(e) Sustained founder (n=34).

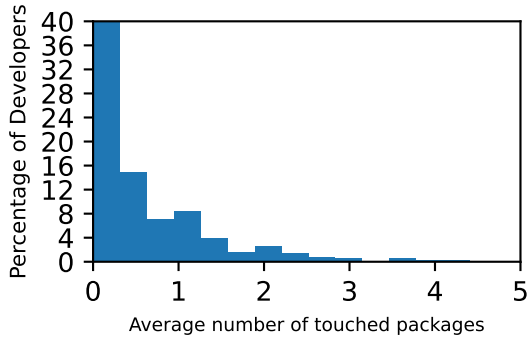


(f) Sustained later joiner (n=203).

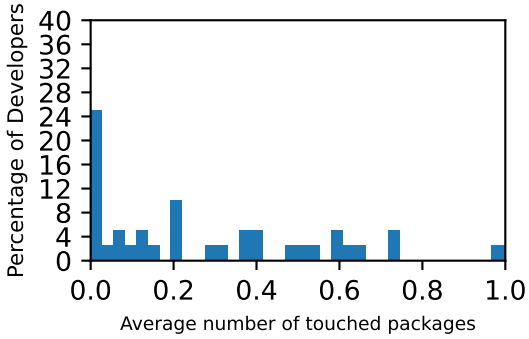
Figure 5: Histogram of the number of developers (y-axis) against the average packages touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



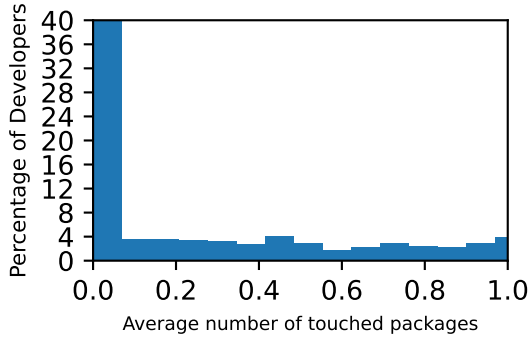
(a) Transient founder (n=28).



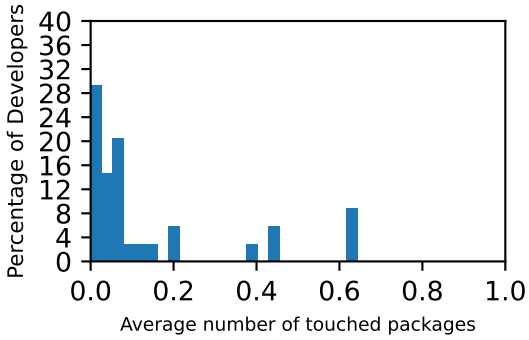
(b) Transient later joiner (n=1396).



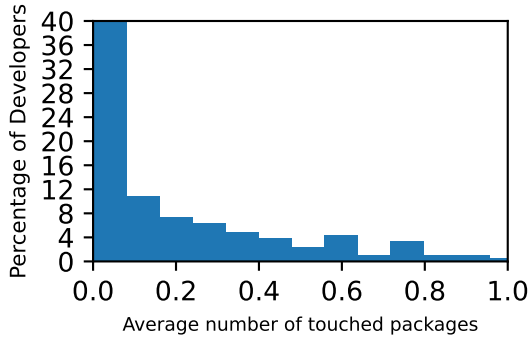
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

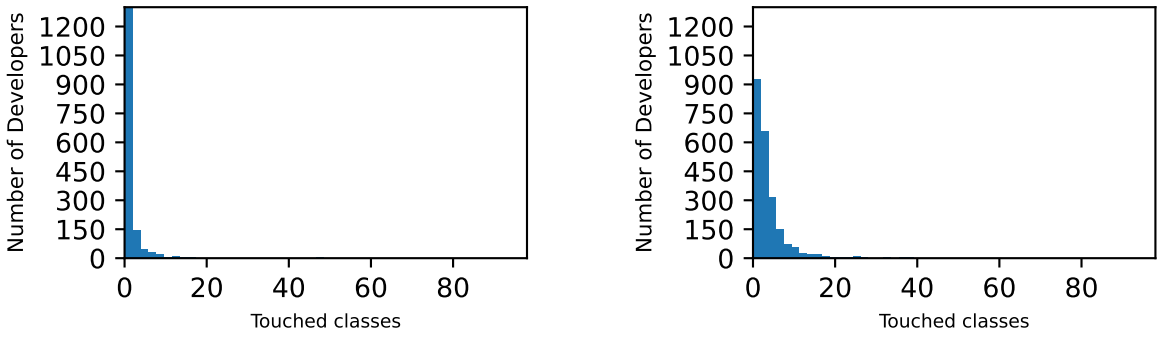


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

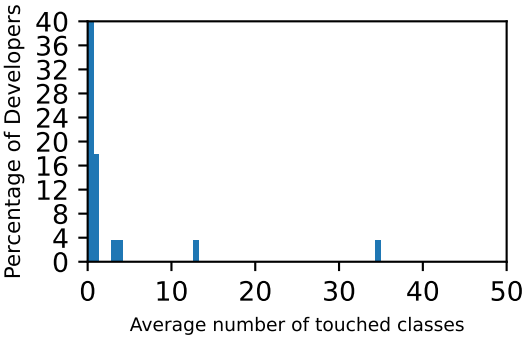
Figure 6: Histogram of the number of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



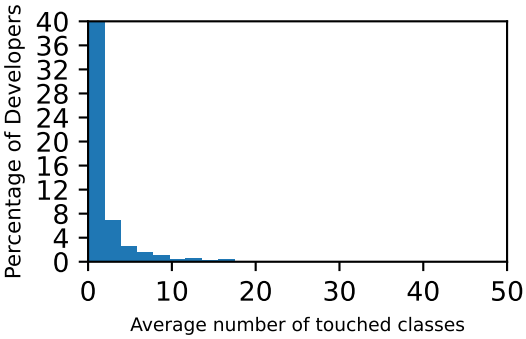
(a) By Day.

(b) By Commit.

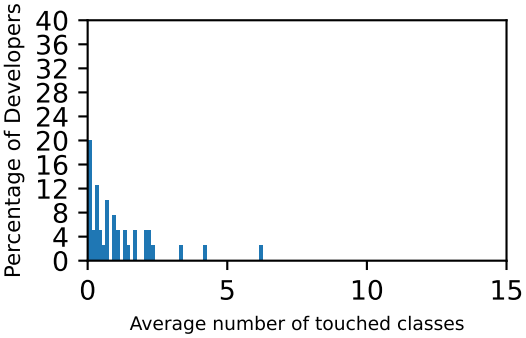
Figure 7: Histogram of the number of developers (y-axis) against the average classes touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



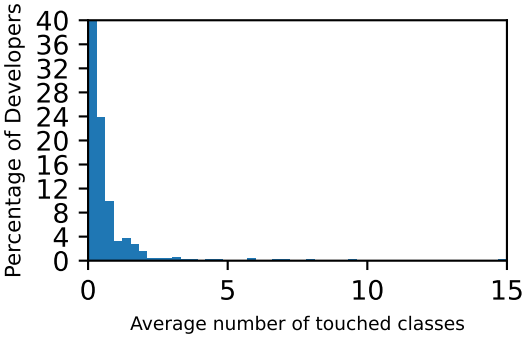
(a) Transient founder (n=28).



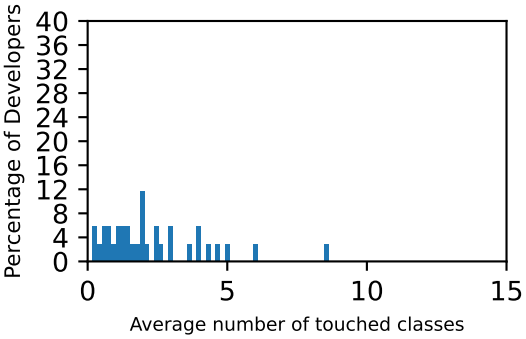
(b) Transient later joiner (n=1396).



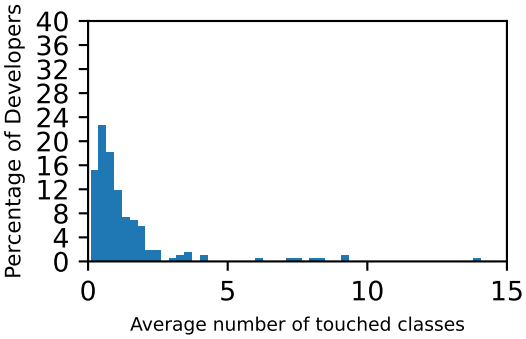
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

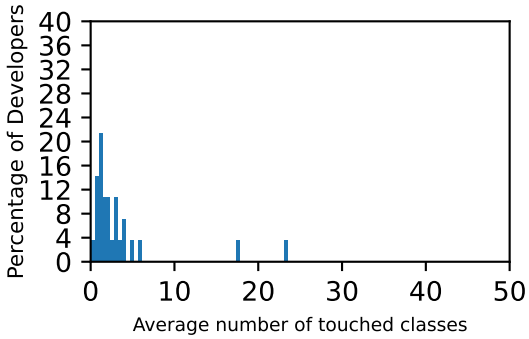


(e) Sustained founder (n=34).

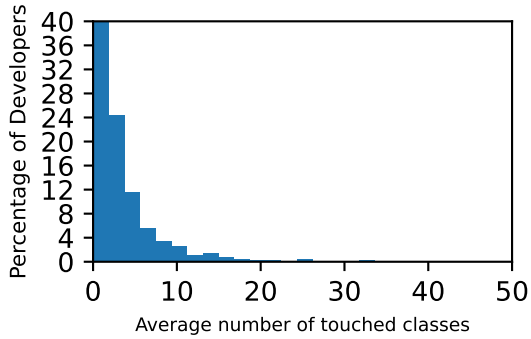


(f) Sustained later joiner (n=203).

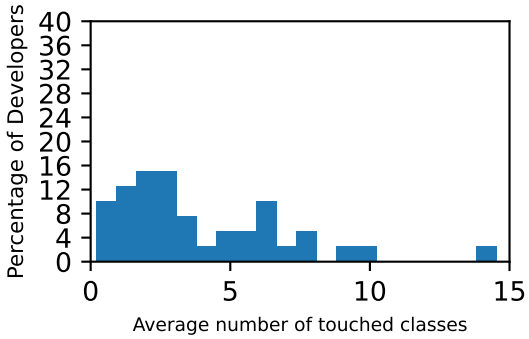
Figure 8: Histogram of the number of developers (y-axis) against the average classes touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



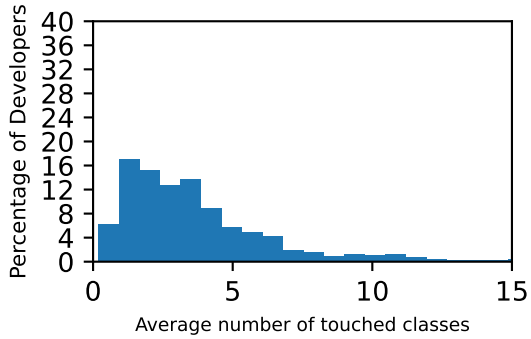
(a) Transient founder (n=28).



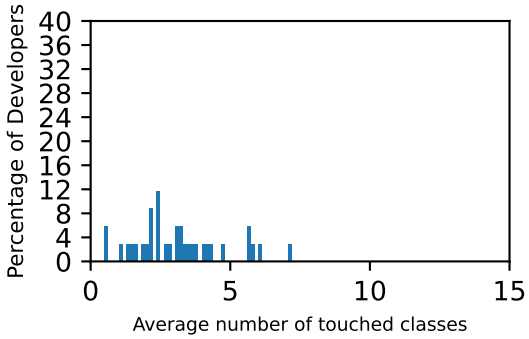
(b) Transient later joiner (n=1396).



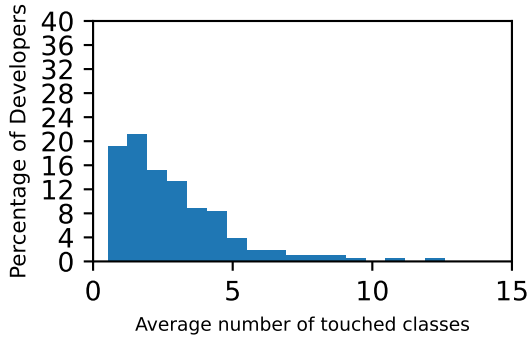
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

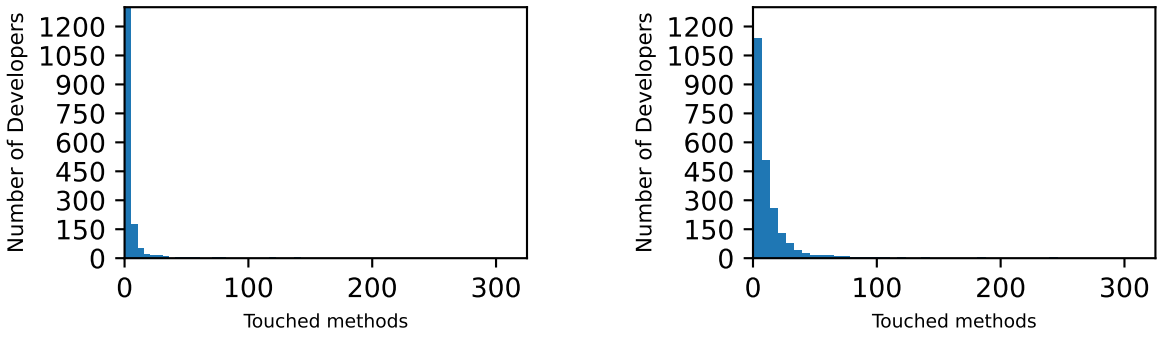


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

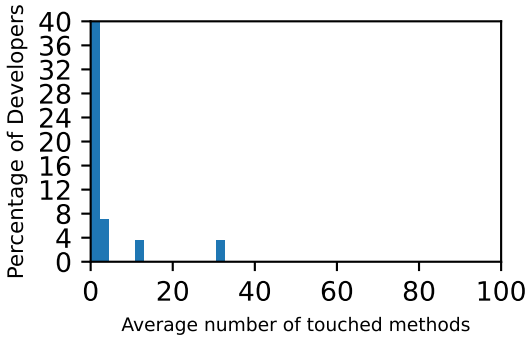
Figure 9: Histogram of the number of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



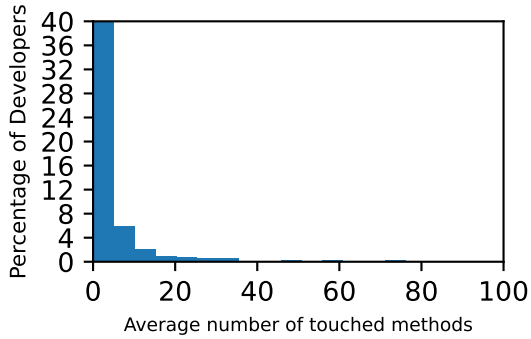
(a) By Day.

(b) By Commit.

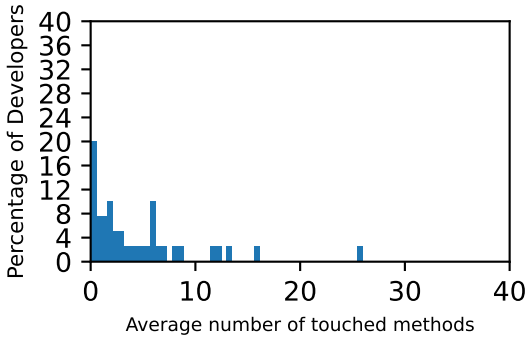
Figure 10: Histogram of the number of developers (y-axis) against the average methods touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



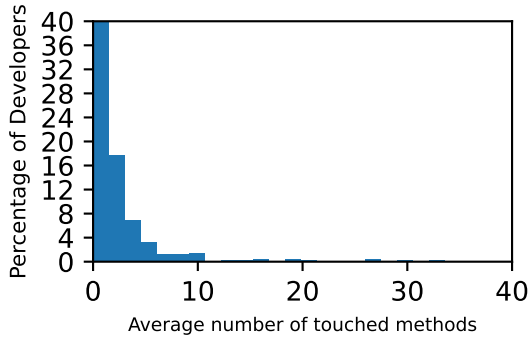
(a) Transient founder (n=28).



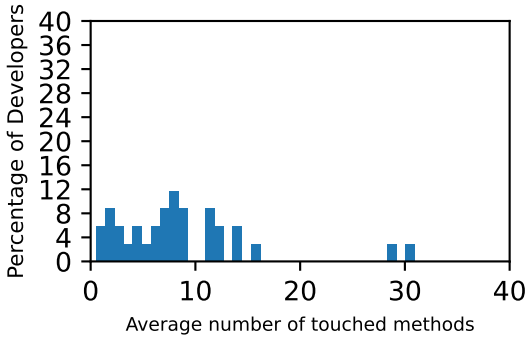
(b) Transient later joiner (n=1396).



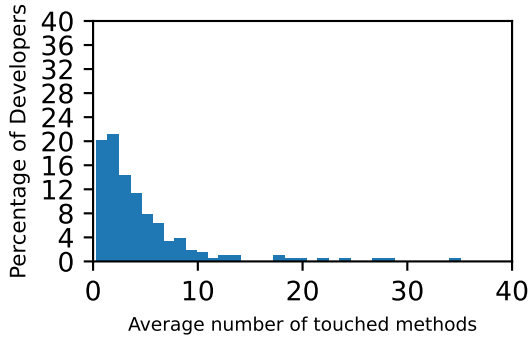
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

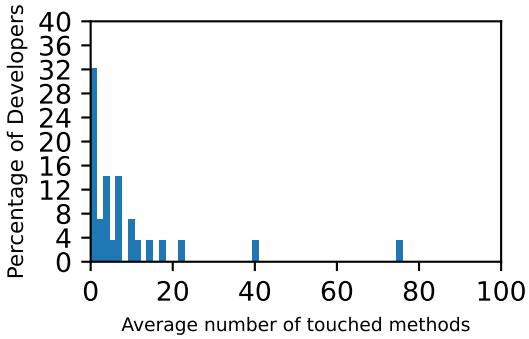


(e) Sustained founder (n=34).

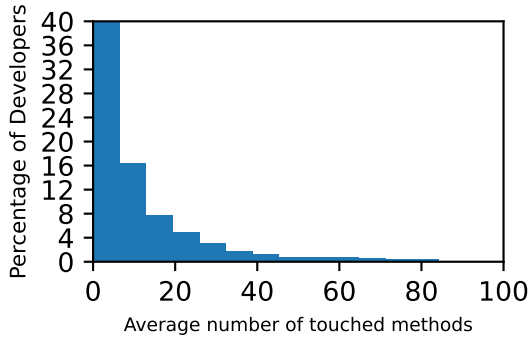


(f) Sustained later joiner (n=203).

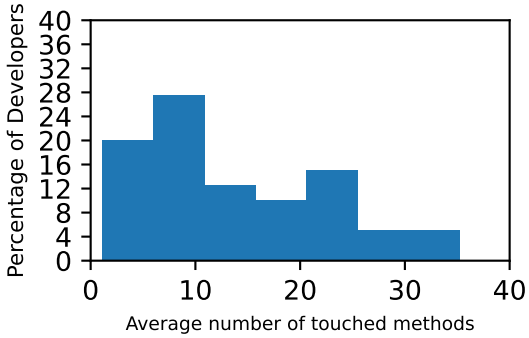
Figure 11: Histogram of the number of developers (y-axis) against the average methods touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



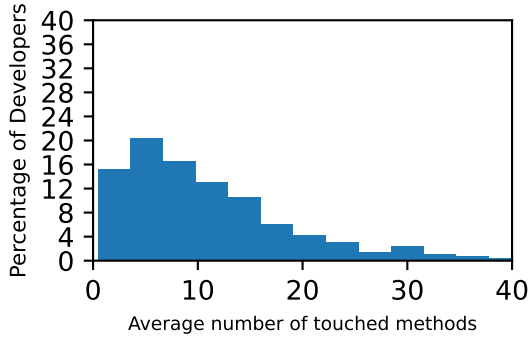
(a) Transient founder (n=28).



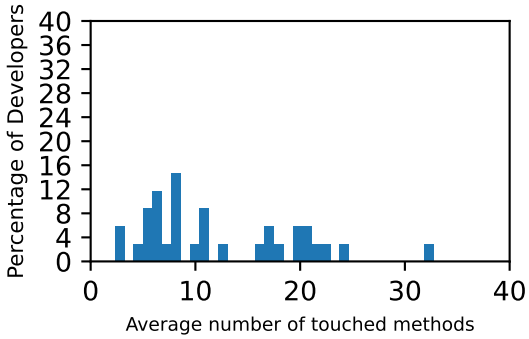
(b) Transient later joiner (n=1396).



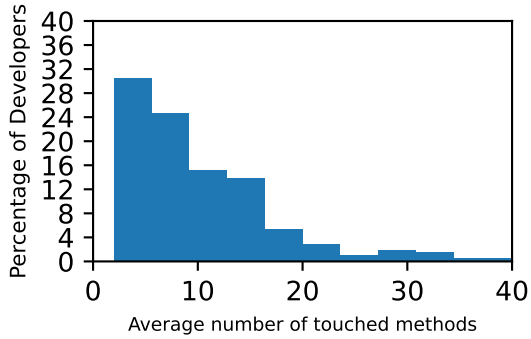
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 12: Histogram of the number of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

1.7 Time series components touched - For all components touched for first five commits and for each period

1.7.1 Time series components touched - For packages touched for first five commits

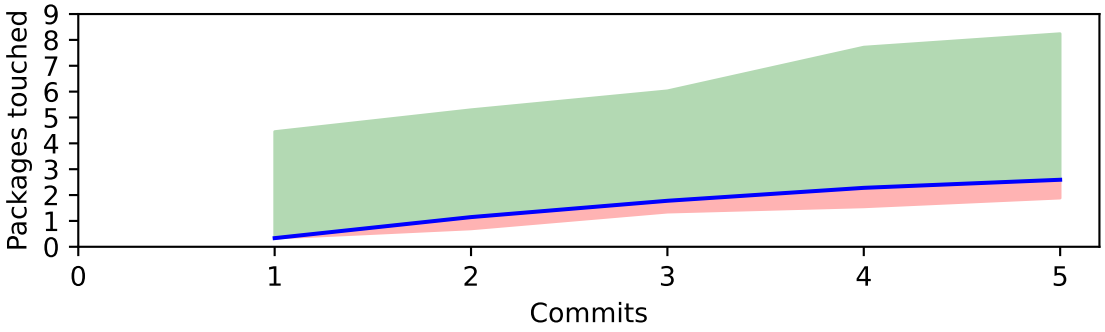


Figure 13: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

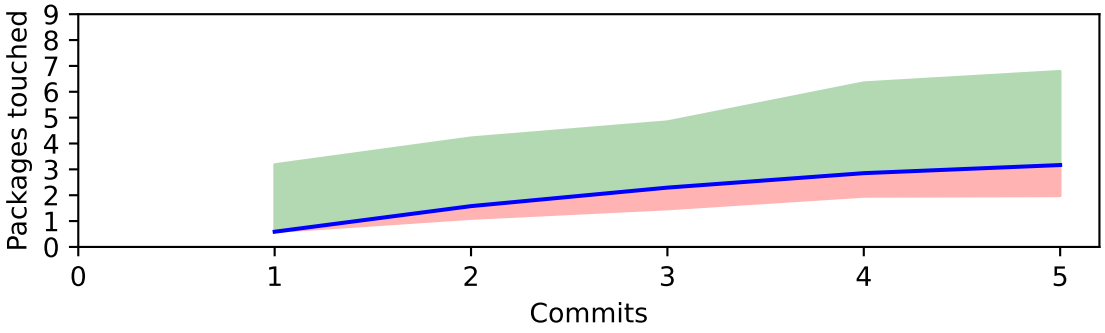


Figure 14: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

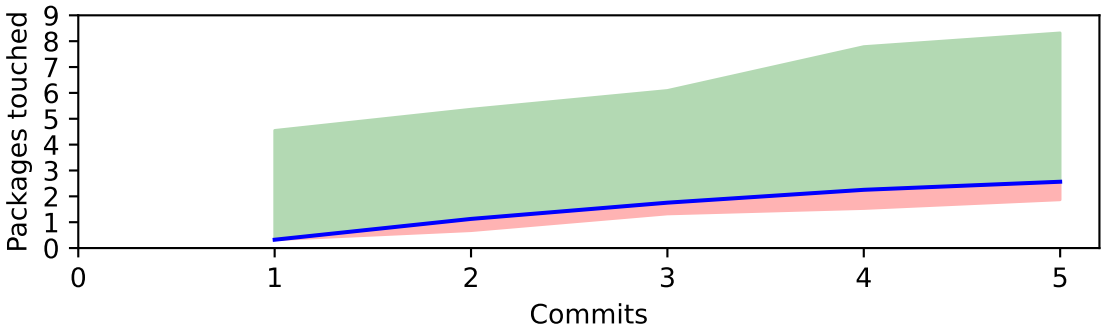


Figure 15: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

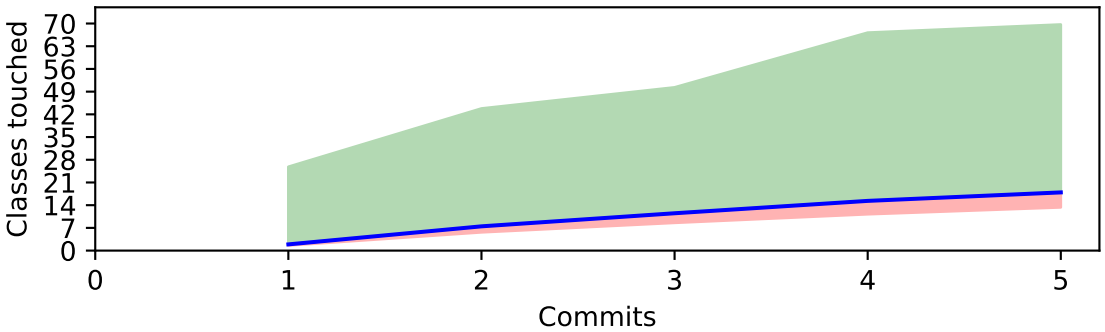


Figure 16: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

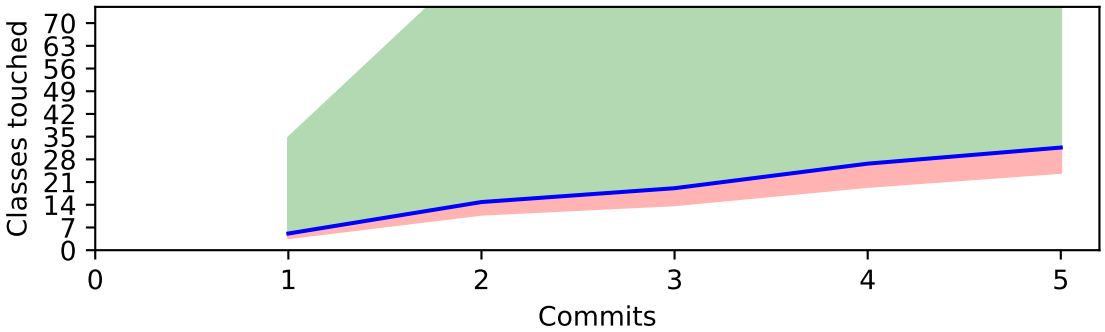


Figure 17: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

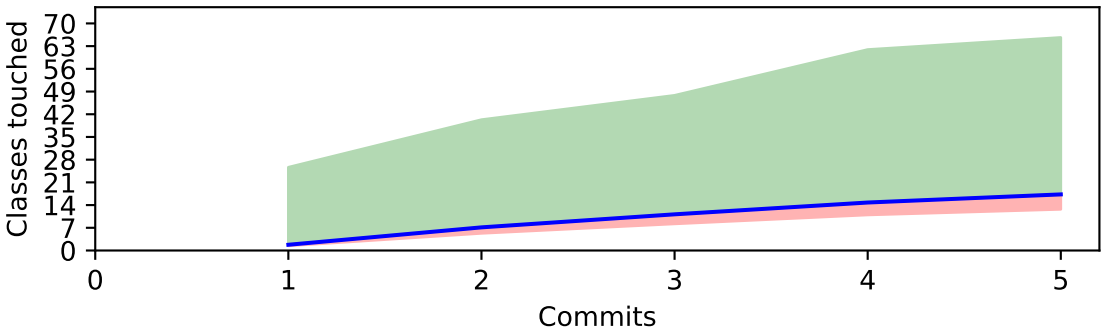


Figure 18: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

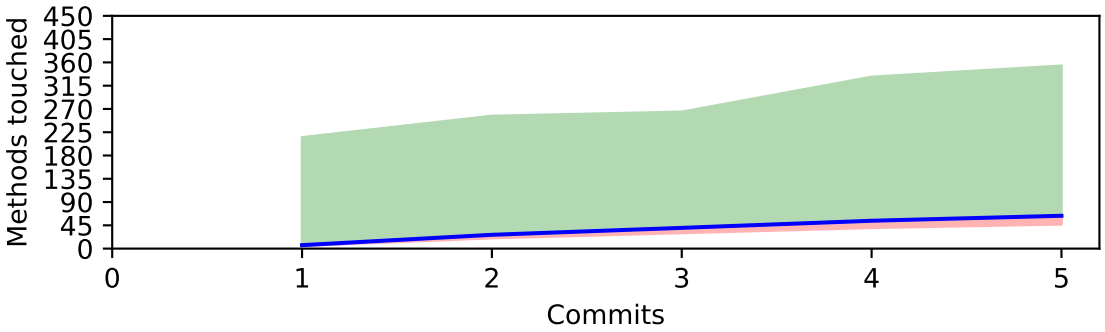


Figure 19: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

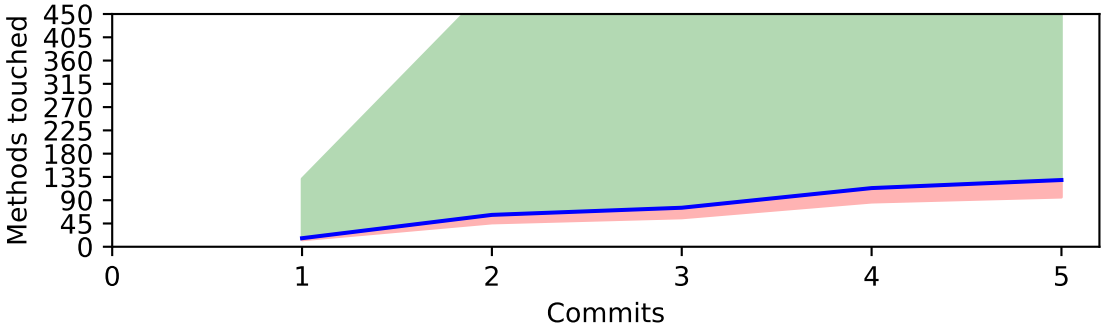


Figure 20: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

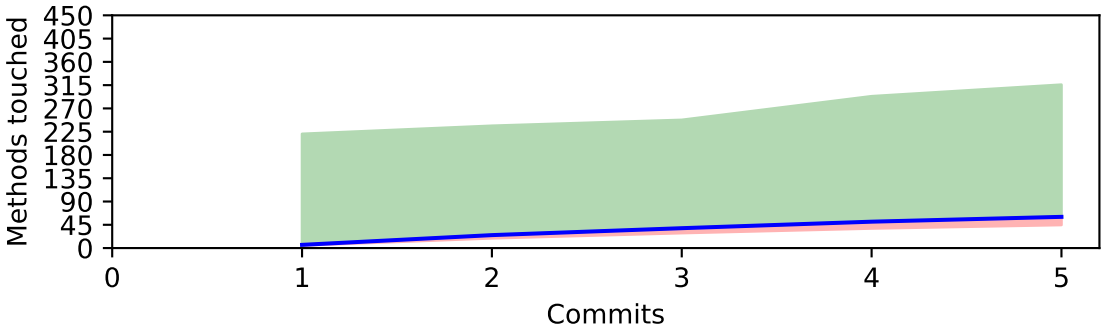
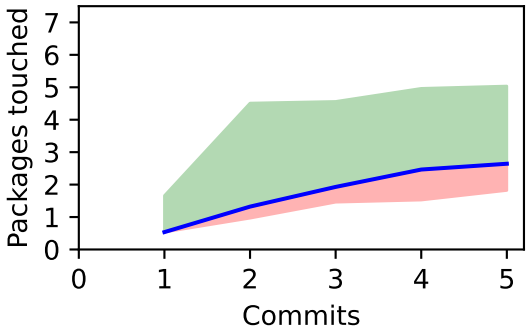


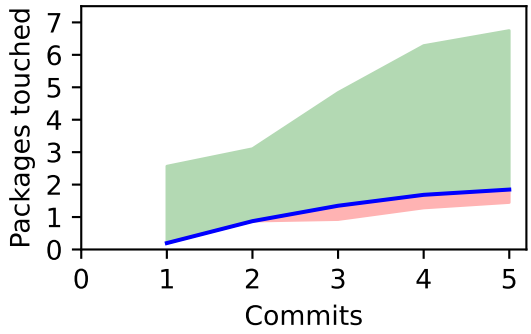
Figure 21: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

1.7.2 Time series components touched - For packages touched for each week first five commits

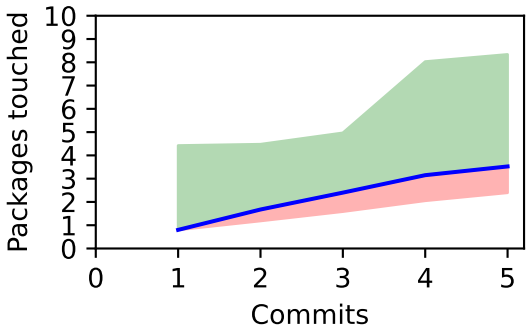
1.7.3 Time series components touched - For packages touched for first ten commits



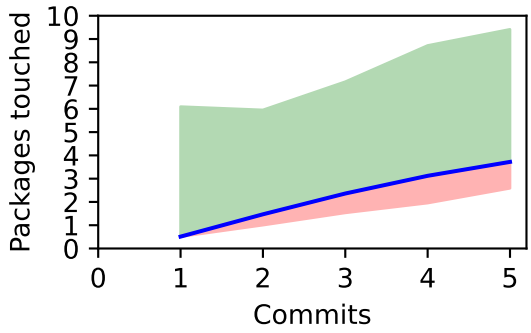
(a) Transient founder (n=28 & μ=2.64)



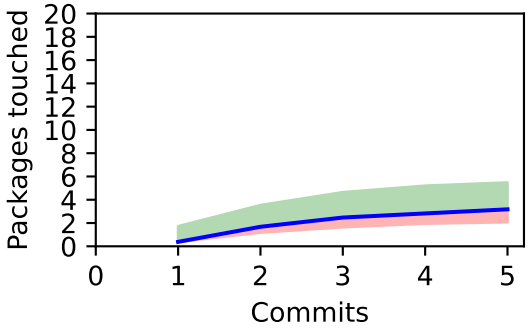
(b) Transient later joiner (n=1396 & μ=1.85)



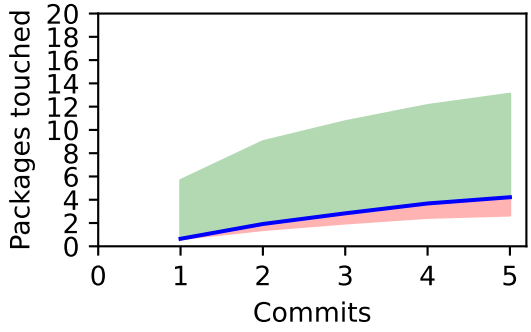
(c) Moderate founder (n=40 & μ=3.52)



(d) Moderate later joiner (n=580 & μ=3.72)

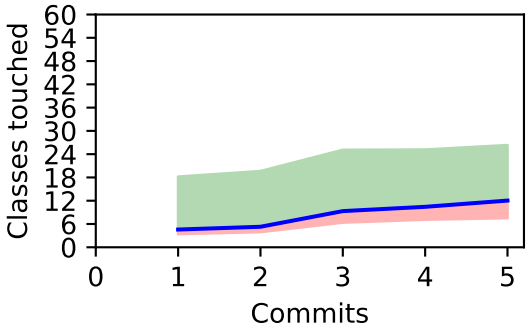


(e) Sustained founder (n=34 & μ=3.18)

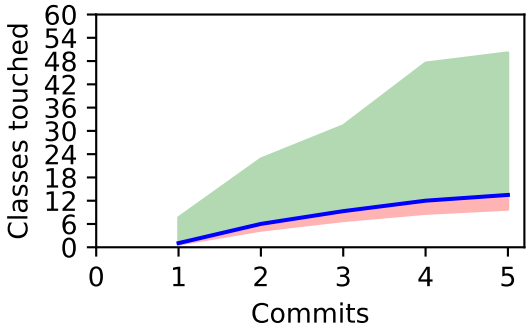


(f) Sustained later joiner (n=203 & μ=4.22)

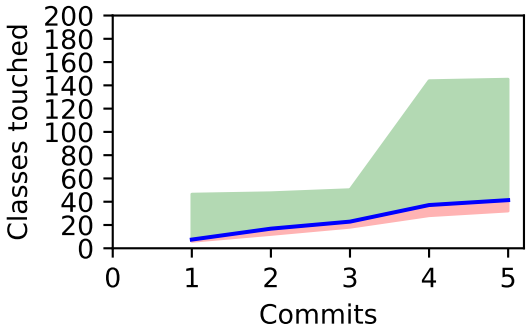
Figure 22: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



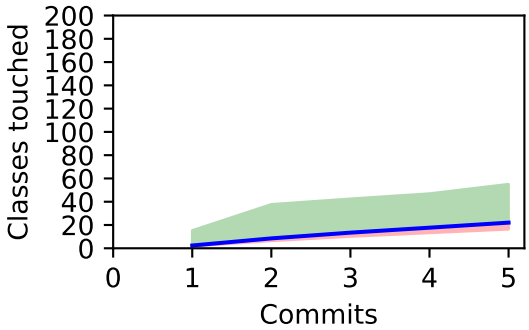
(a) Transient founder (n=28 & μ=12.04)



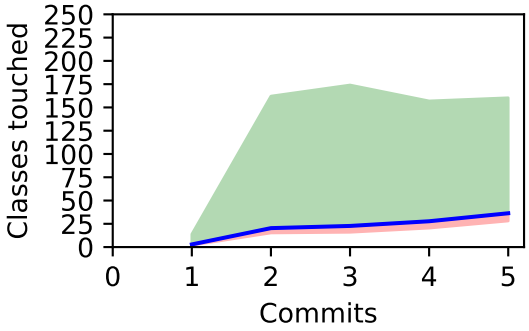
(b) Transient later joiner (n=1396 & μ=13.49)



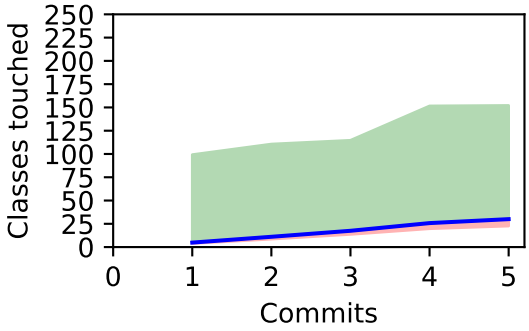
(c) Moderate founder (n=40 & μ=41.38)



(d) Moderate later joiner (n=580 & μ=22.04)

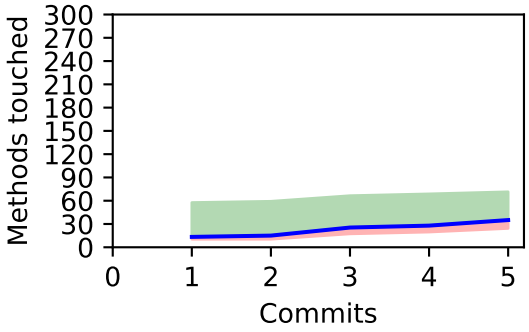


(e) Sustained founder (n=34 & μ=36.32)

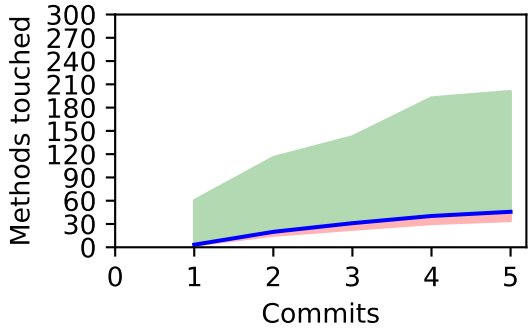


(f) Sustained later joiner (n=203 & μ=30.01)

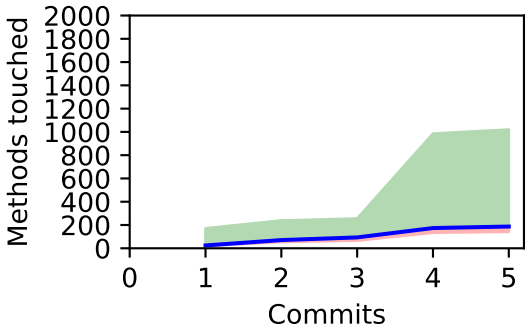
Figure 23: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



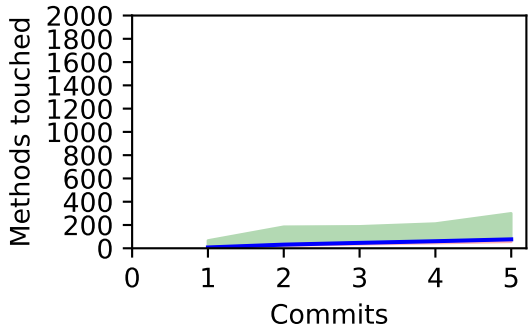
(a) Transient founder (n=28 & μ=35.07)



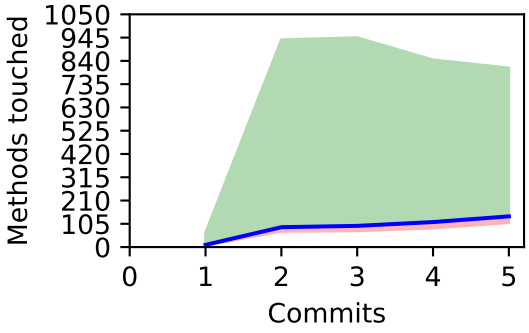
(b) Transient later joiner (n=1396 & μ=45.72)



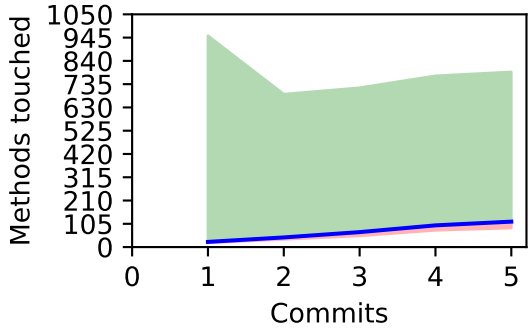
(c) Moderate founder (n=40 & μ=186.75)



(d) Moderate later joiner (n=580 & μ=76.85)



(e) Sustained founder (n=34 & μ=138.53)



(f) Sustained later joiner (n=203 & μ=114.77)

Figure 24: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

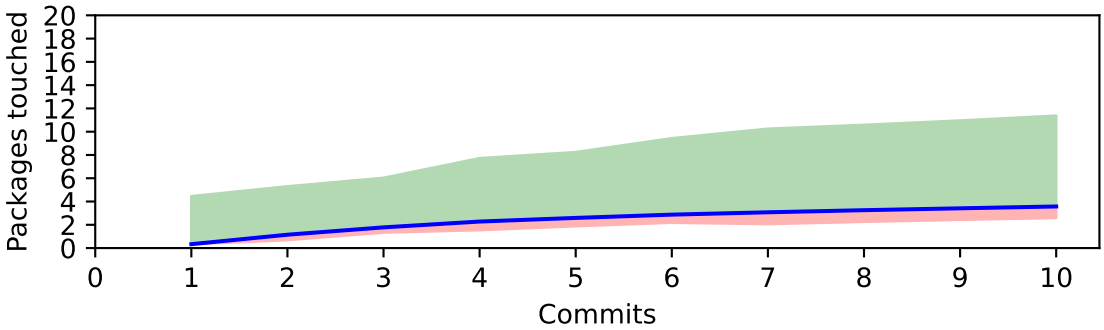


Figure 25: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

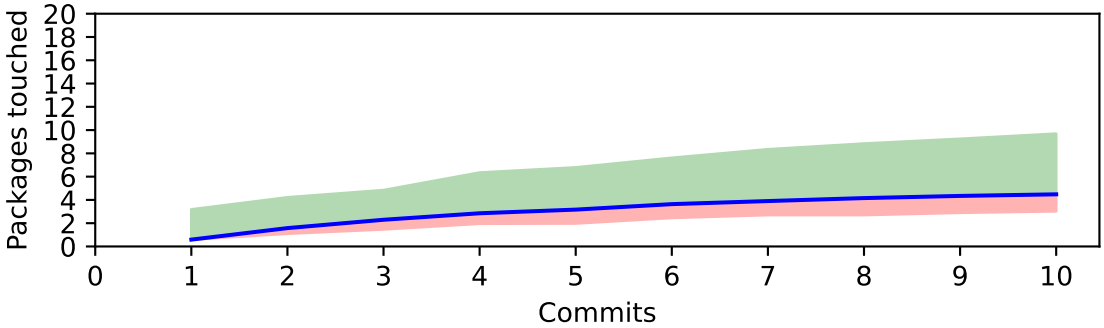


Figure 26: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

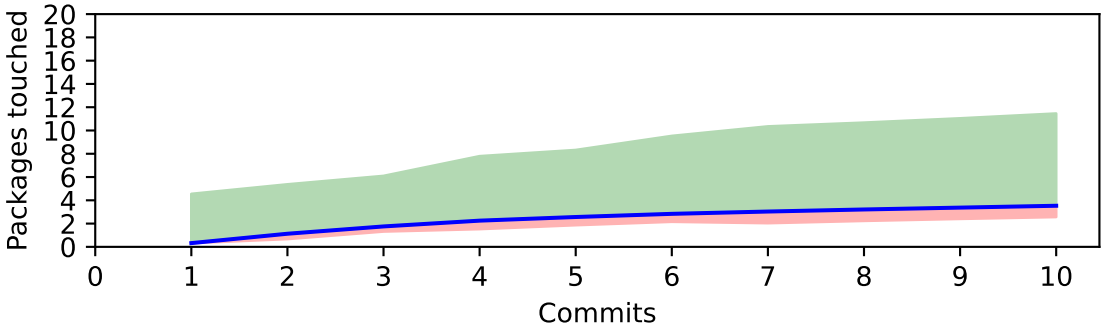


Figure 27: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

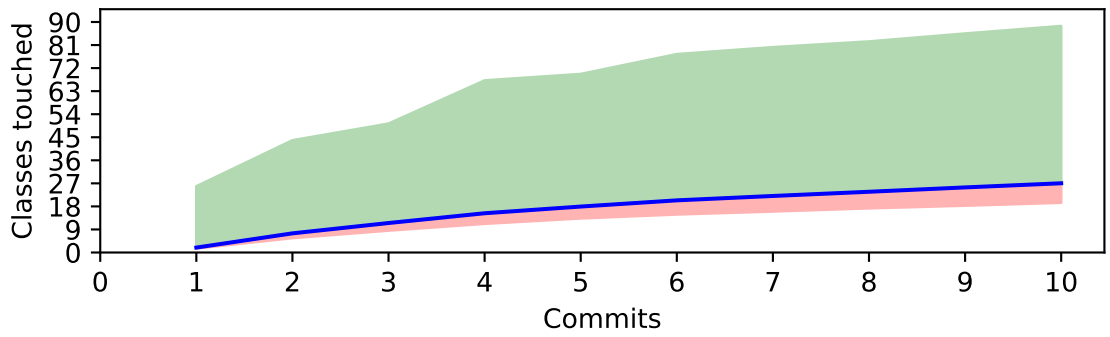


Figure 28: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

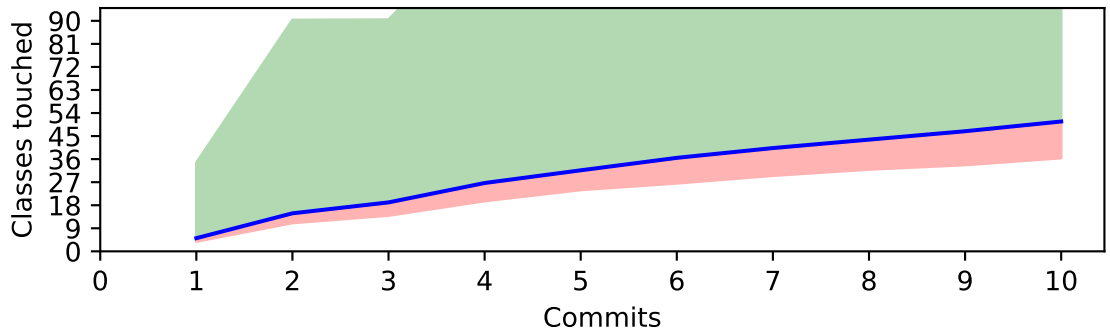


Figure 29: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

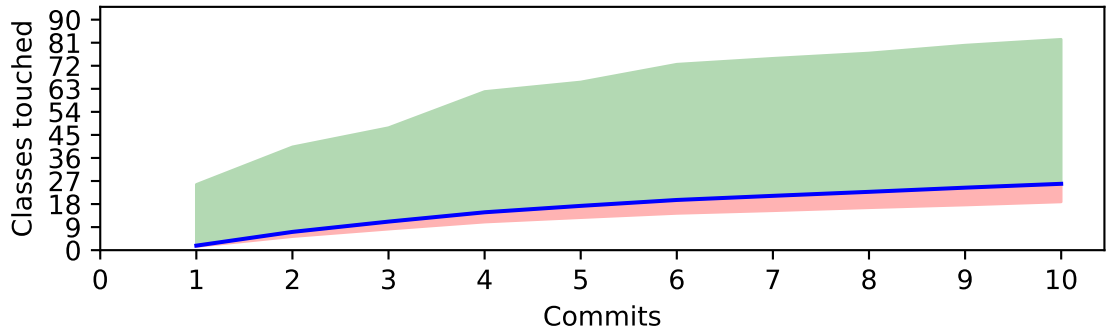


Figure 30: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

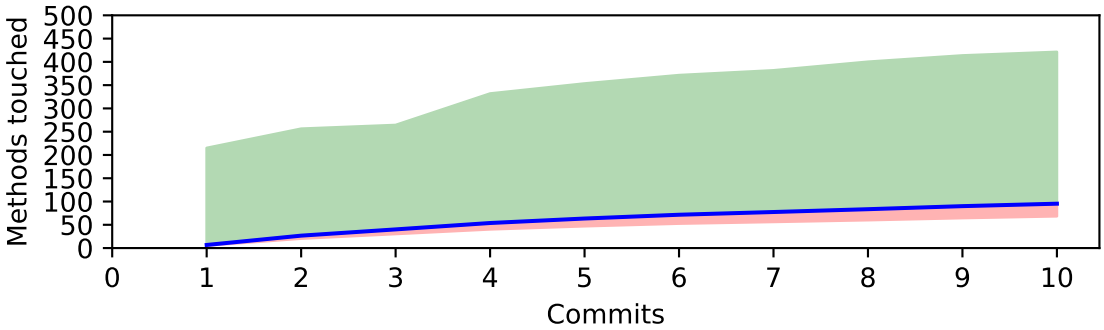


Figure 31: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

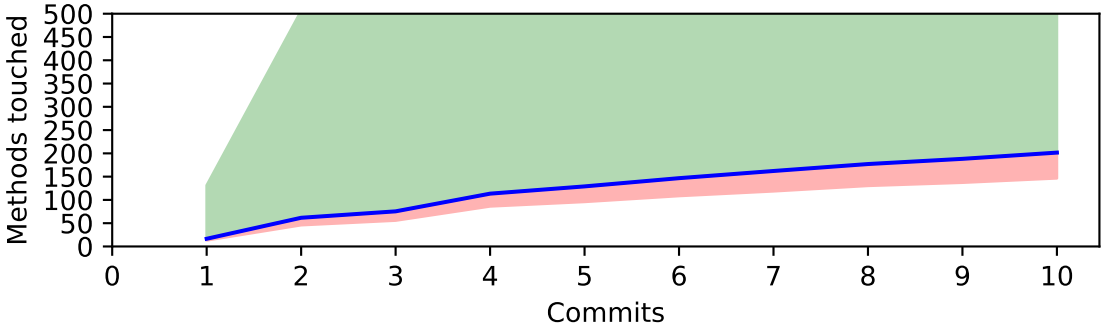


Figure 32: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

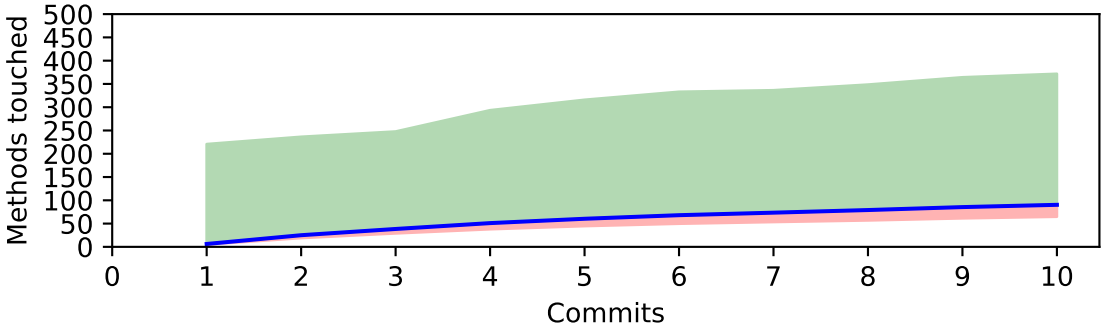
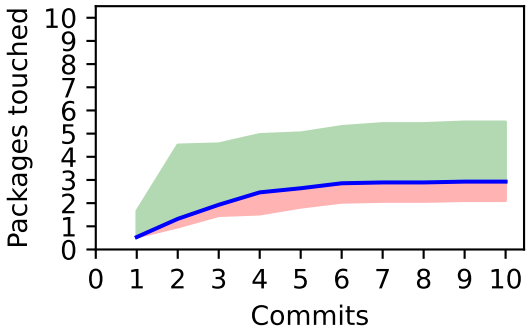


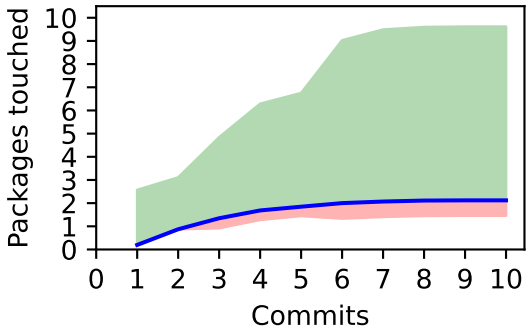
Figure 33: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

1.7.4 Time series components touched - For packages touched for each week first ten commits

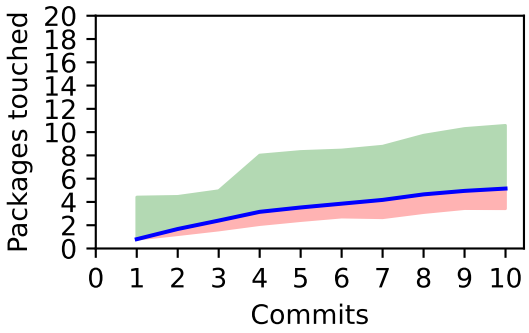
1.7.5 Time series components touched - For packages touched for first twenty commits



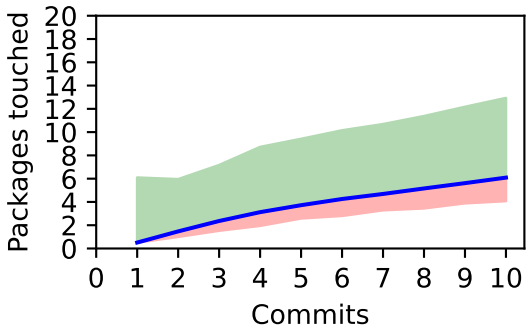
(a) Transient founder (n=28 & μ=2.93)



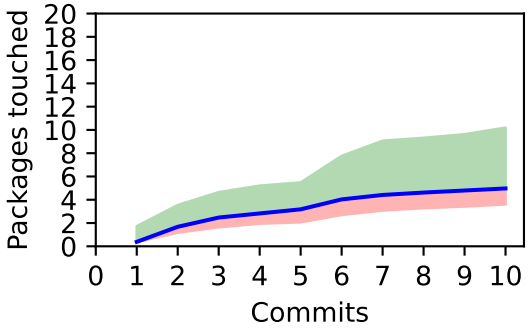
(b) Transient later joiner (n=1396 & μ=2.12)



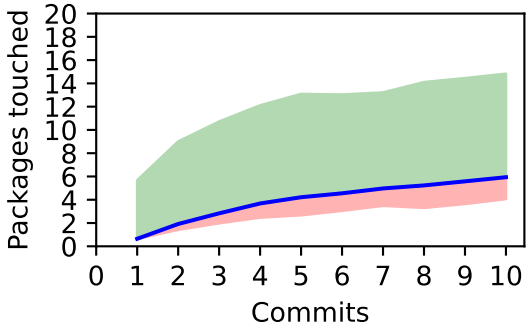
(c) Moderate founder (n=40 & μ=5.15)



(d) Moderate later joiner (n=580 & μ=6.09)

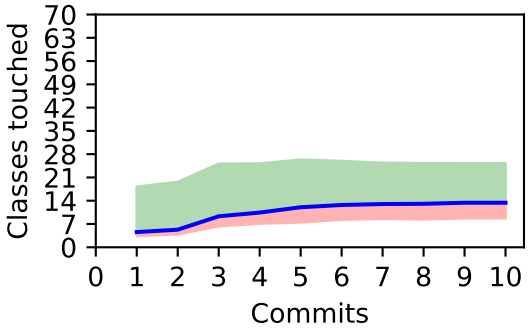


(e) Sustained founder (n=34 & μ=4.97)

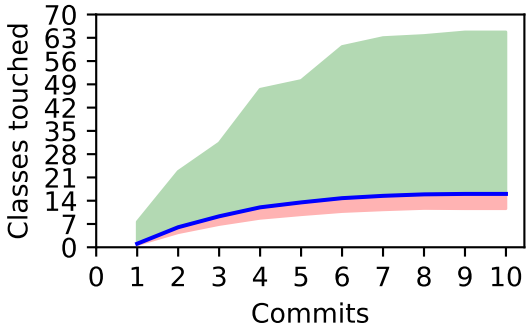


(f) Sustained later joiner (n=203 & μ=5.94)

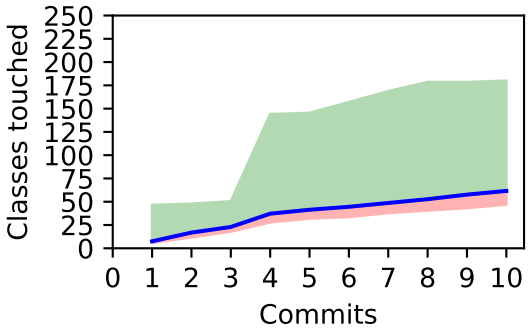
Figure 34: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



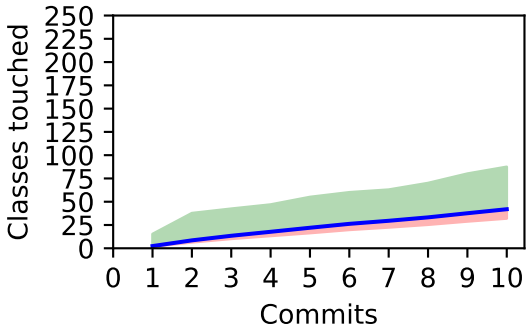
(a) Transient founder (n=28 & μ=13.39)



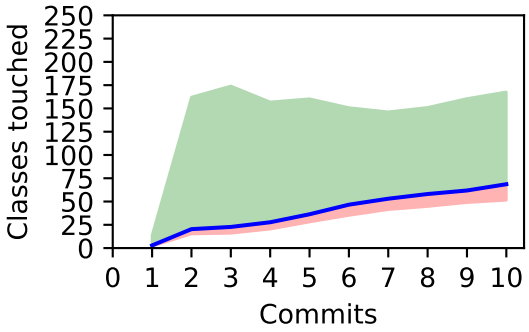
(b) Transient later joiner (n=1396 & μ=16.05)



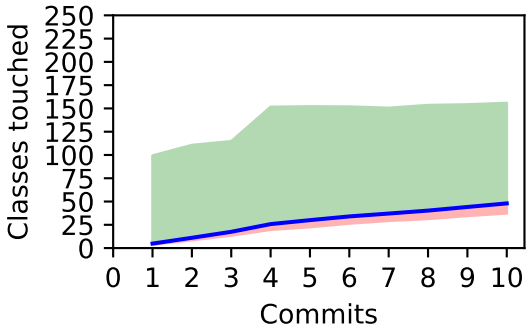
(c) Moderate founder (n=40 & μ=61.6)



(d) Moderate later joiner (n=580 & μ=41.95)

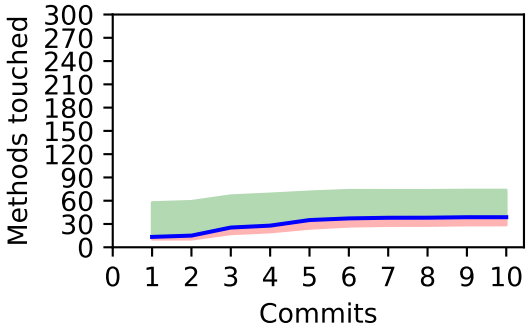


(e) Sustained founder (n=34 & μ=68.59)

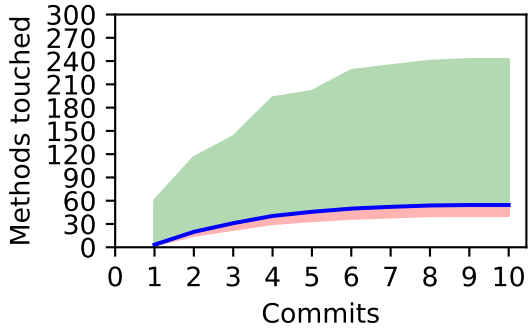


(f) Sustained later joiner (n=203 & μ=47.98)

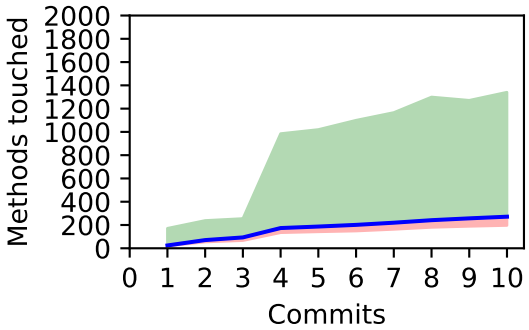
Figure 35: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



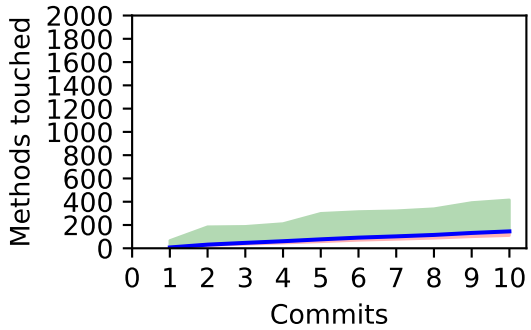
(a) Transient founder (n=28 & μ=38.75)



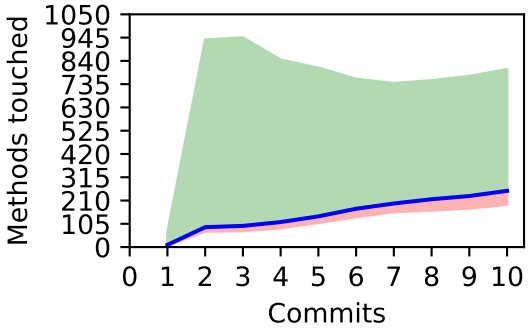
(b) Transient later joiner (n=1396 & μ=54.43)



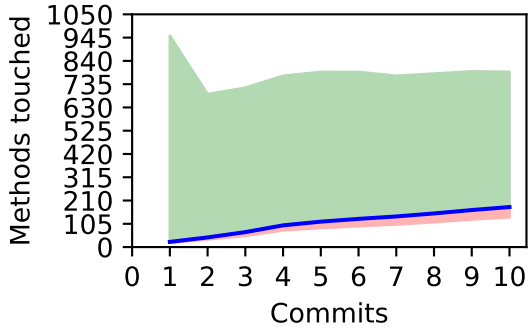
(c) Moderate founder (n=40 & μ=271.42)



(d) Moderate later joiner (n=580 & μ=145.21)



(e) Sustained founder (n=34 & μ=253.65)



(f) Sustained later joiner (n=203 & μ=180.7)

Figure 36: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

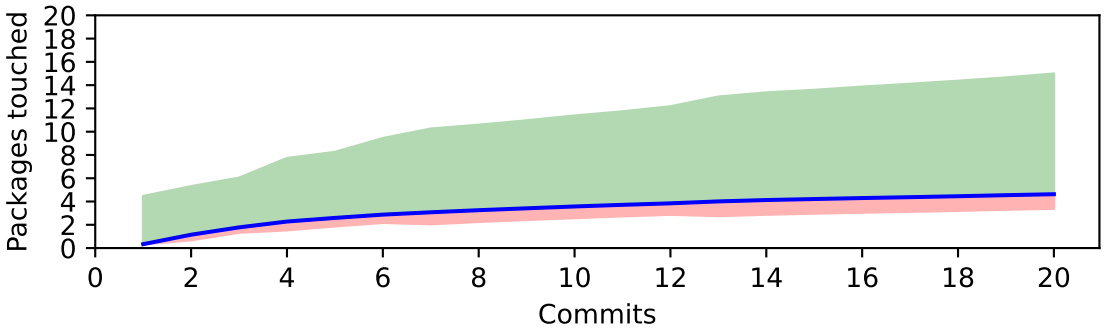


Figure 37: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

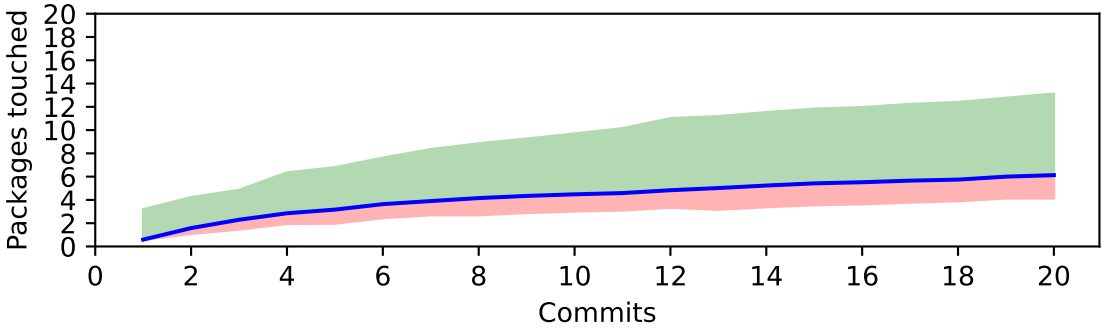


Figure 38: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

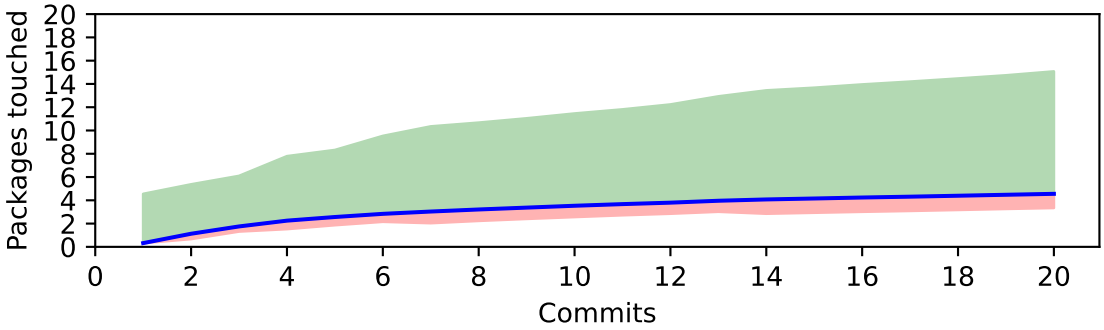


Figure 39: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

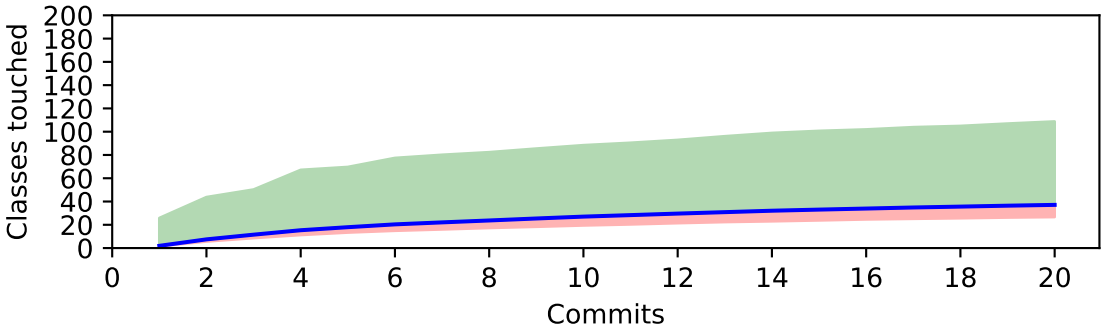


Figure 40: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

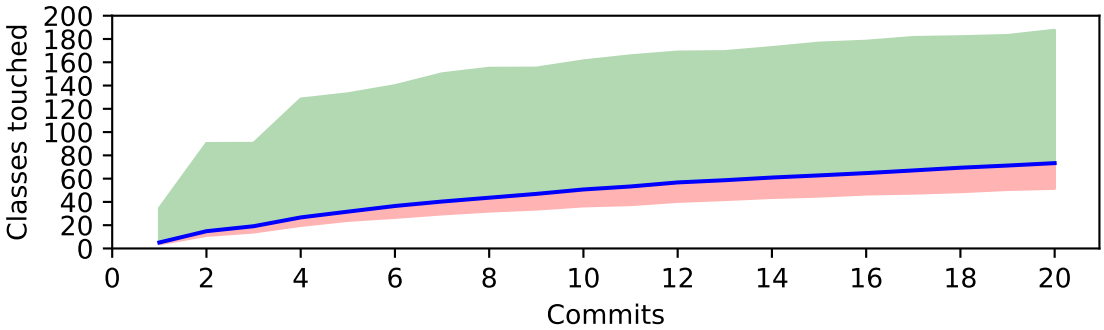


Figure 41: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

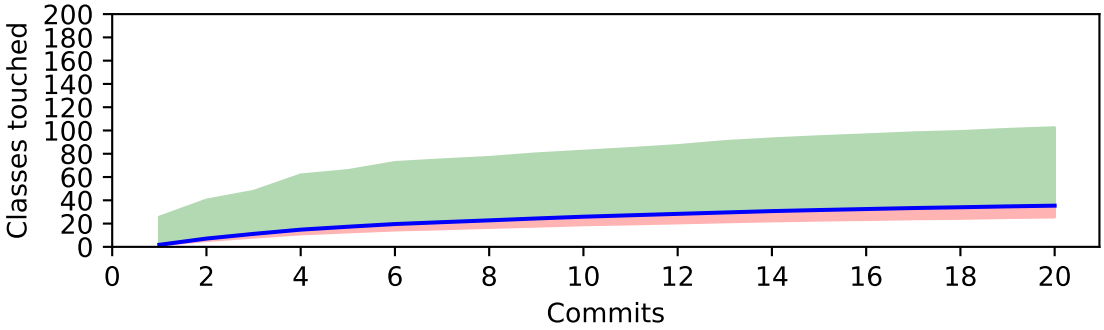


Figure 42: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

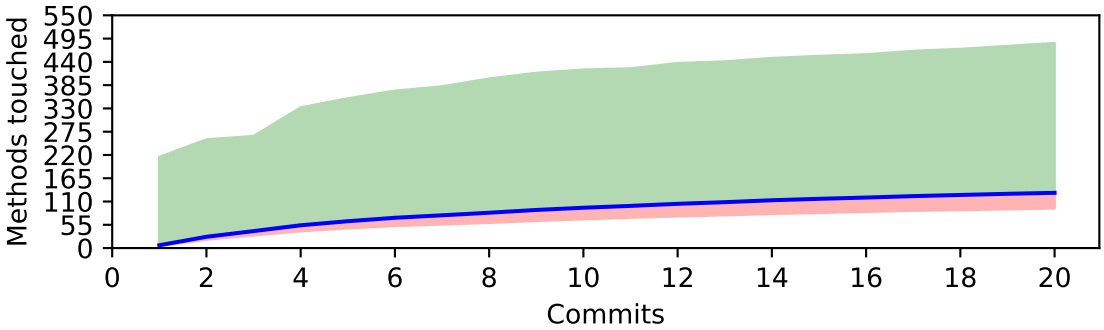


Figure 43: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

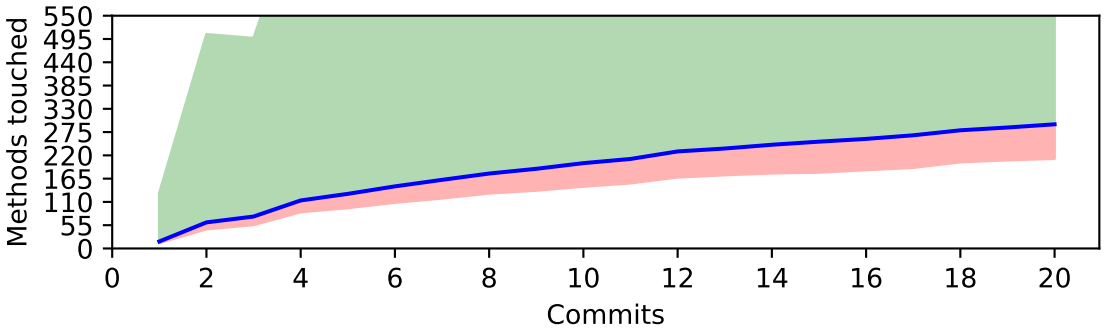


Figure 44: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

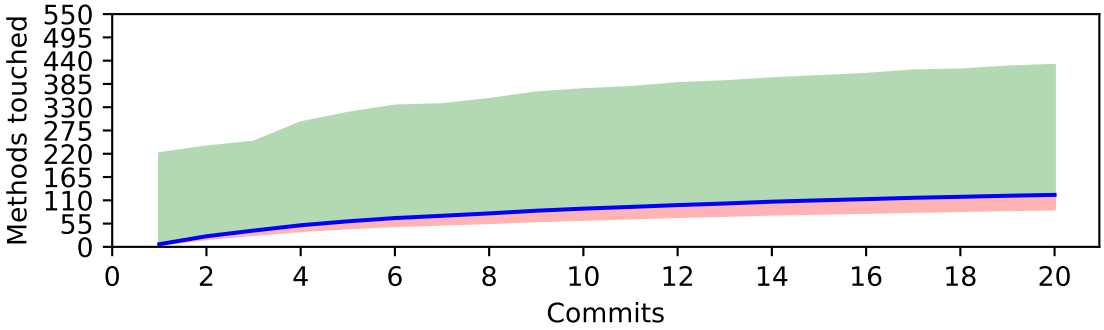
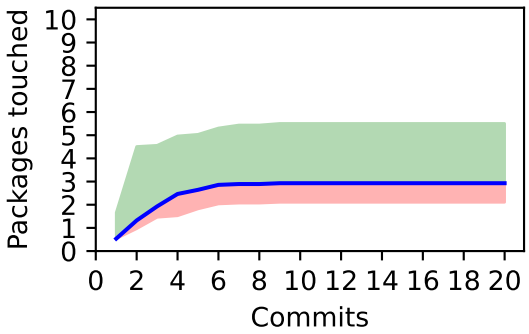


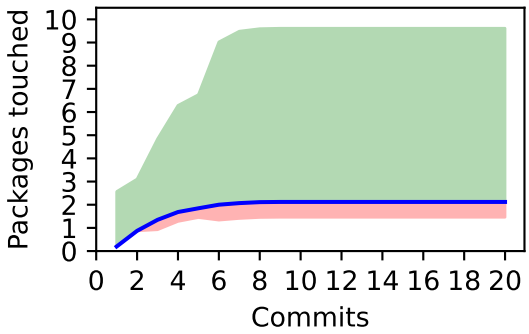
Figure 45: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

1.7.6 Time series components touched - For packages touched for each week first twenty commits

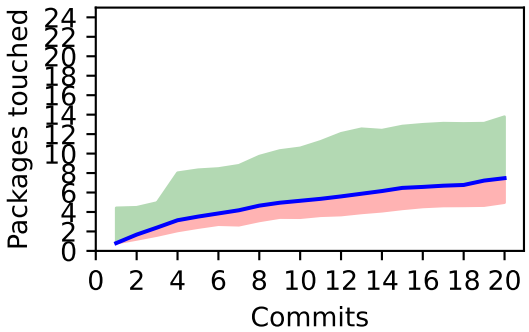
1.7.7 Time series components touched - For packages touched for commit



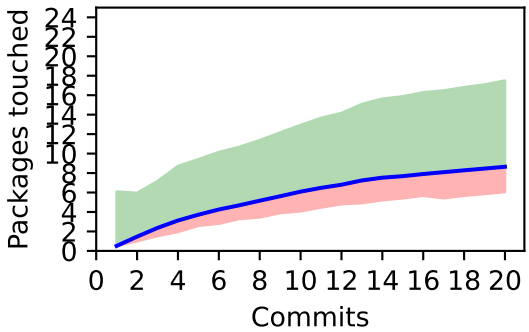
(a) Transient founder (n=28 & μ=2.93)



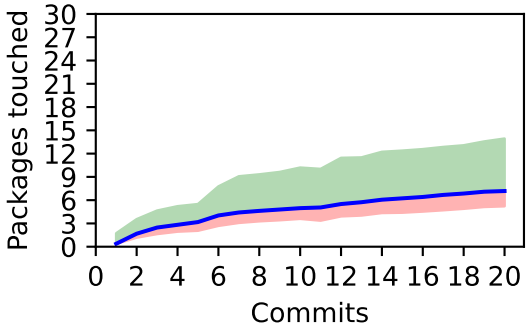
(b) Transient later joiner (n=1396 & μ=2.12)



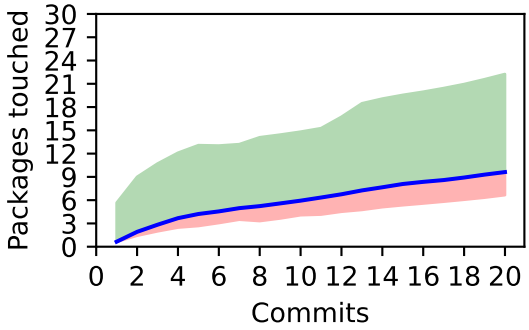
(c) Moderate founder (n=40 & μ=7.48)



(d) Moderate later joiner (n=580 & μ=8.65)

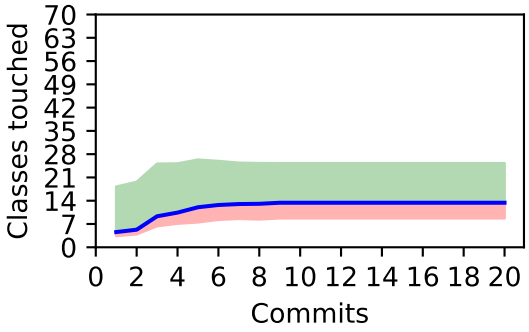


(e) Sustained founder (n=34 & μ=7.18)

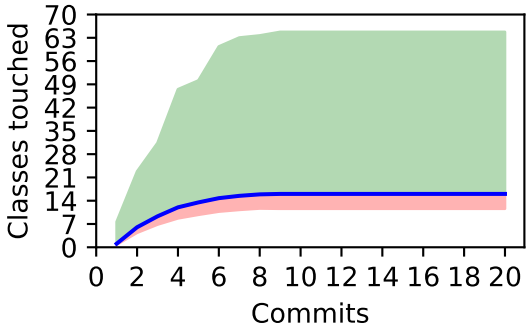


(f) Sustained later joiner (n=203 & μ=9.62)

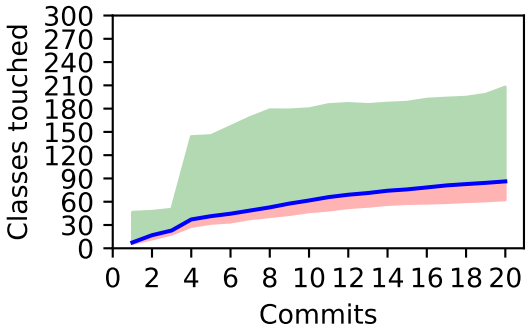
Figure 46: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



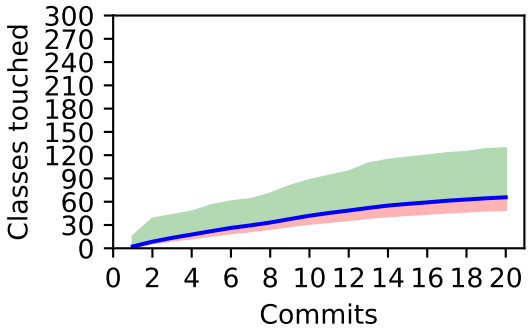
(a) Transient founder (n=28 & μ=13.39)



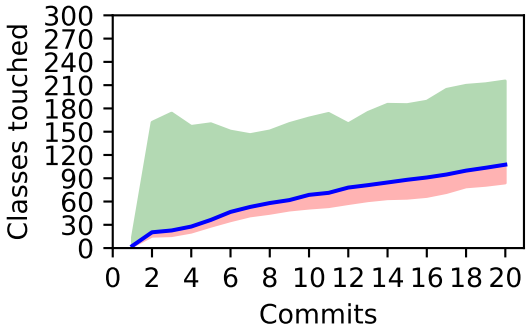
(b) Transient later joiner (n=1396 & μ=16.05)



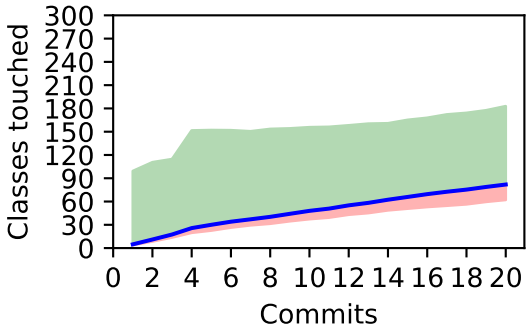
(c) Moderate founder (n=40 & μ=86.22)



(d) Moderate later joiner (n=580 & μ=65.7)

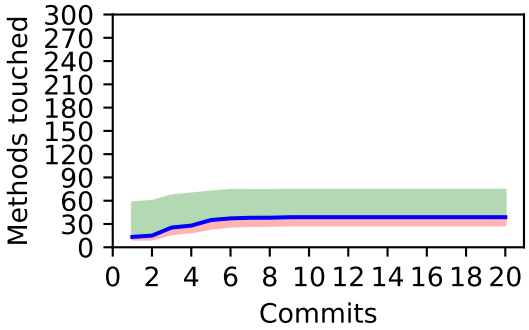


(e) Sustained founder (n=34 & μ=107.5)

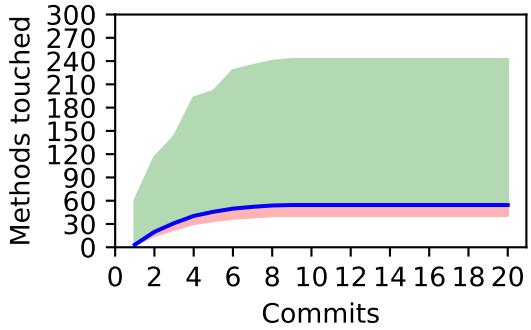


(f) Sustained later joiner (n=203 & μ=81.91)

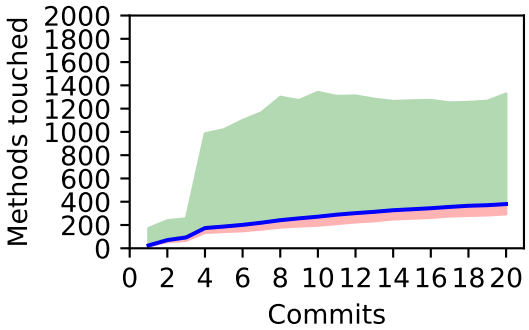
Figure 47: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



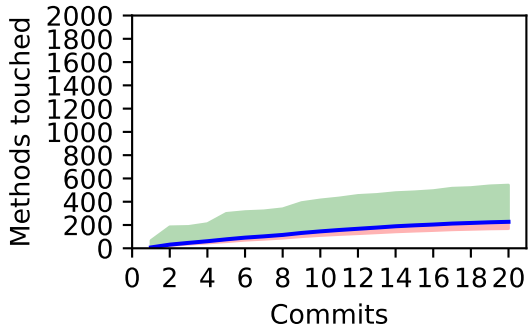
(a) Transient founder (n=28 & μ=38.75)



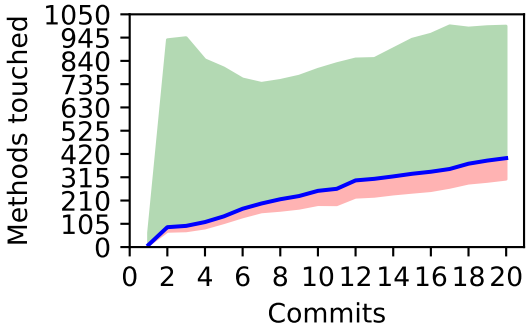
(b) Transient later joiner (n=1396 & μ=54.43)



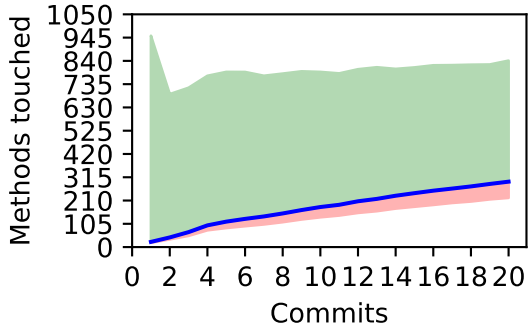
(c) Moderate founder (n=40 & μ=379.9)



(d) Moderate later joiner (n=580 & μ=227.38)



(e) Sustained founder (n=34 & μ=401.24)



(f) Sustained later joiner (n=203 & μ=295.1)

Figure 48: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

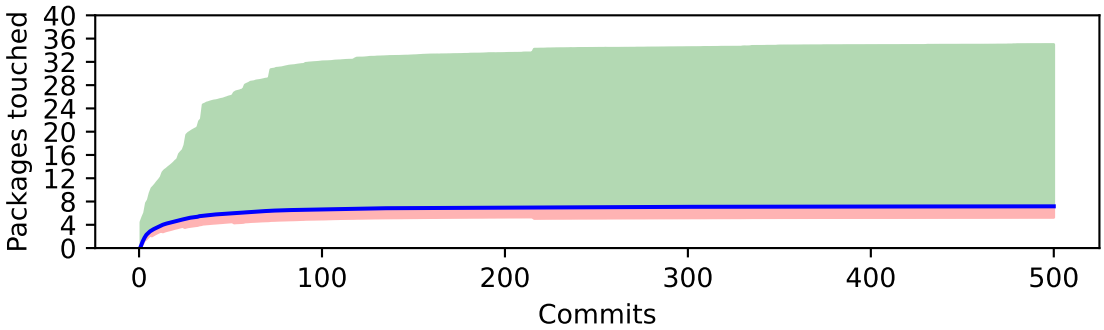


Figure 49: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

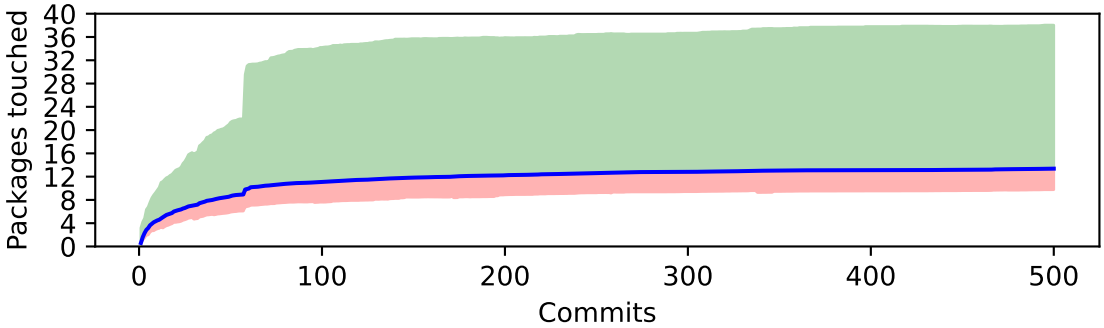


Figure 50: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

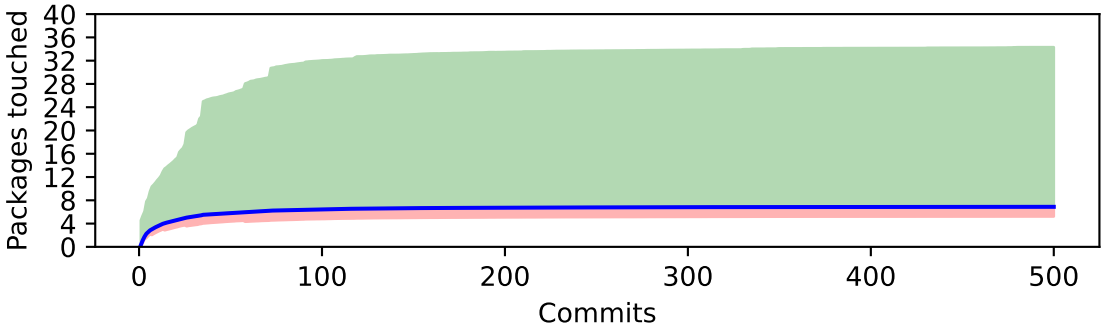


Figure 51: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

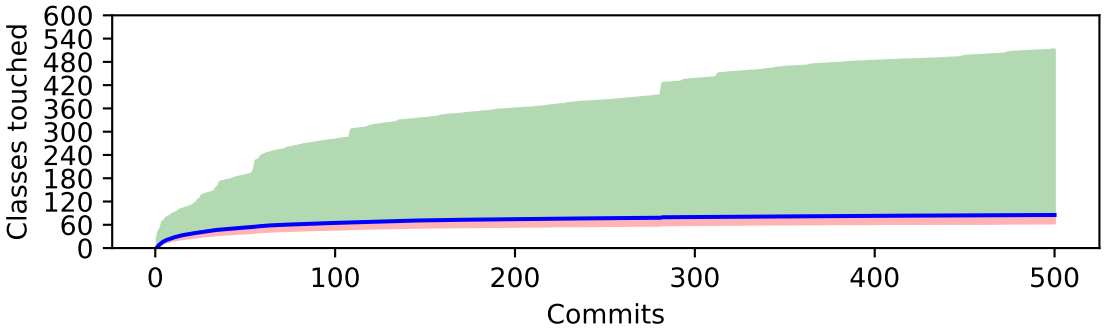


Figure 52: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

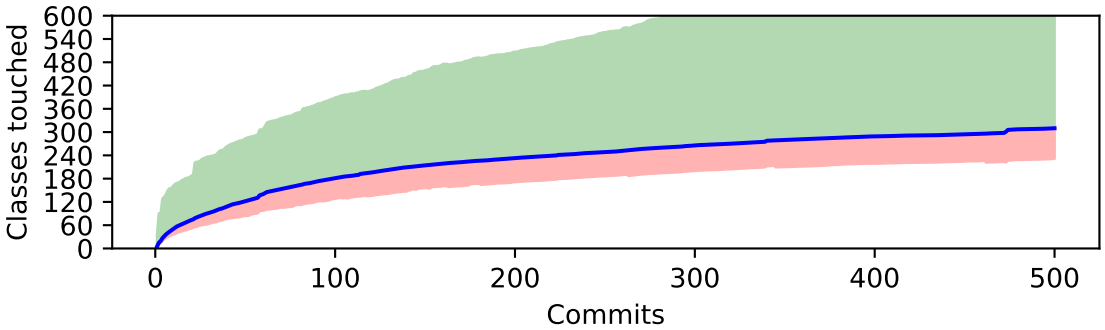


Figure 53: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

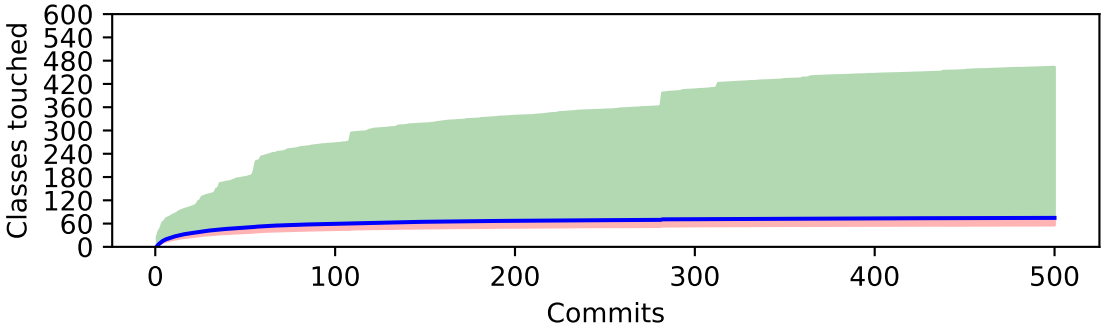


Figure 54: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

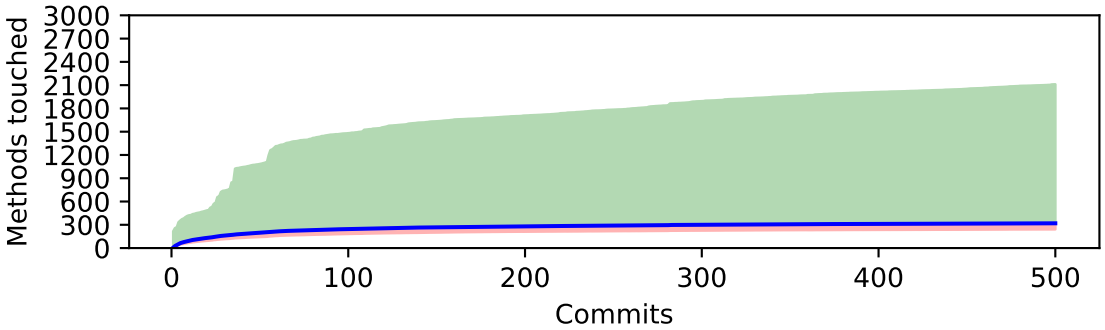


Figure 55: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

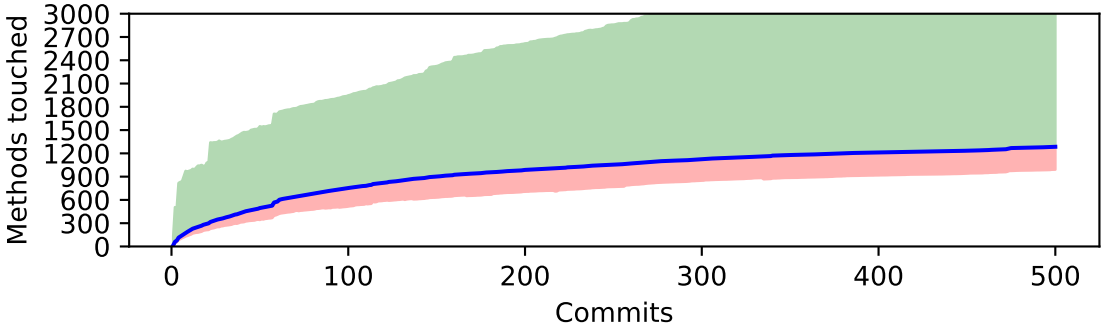


Figure 56: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

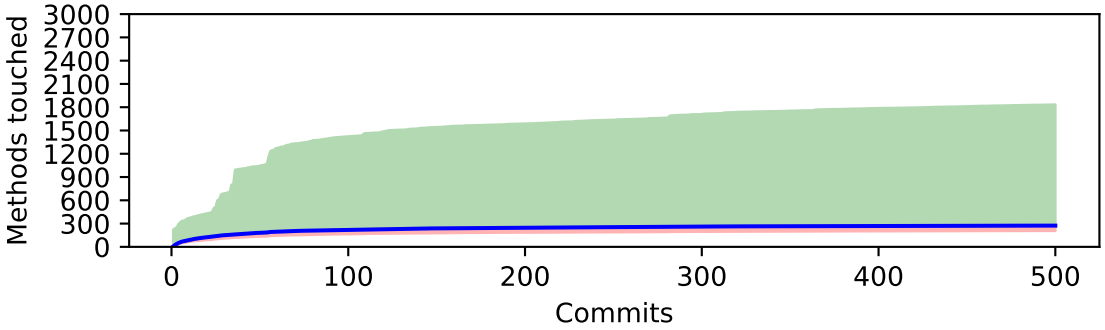
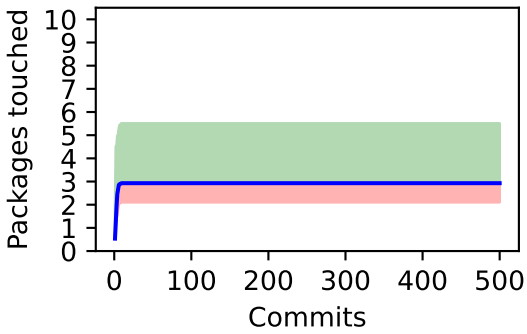
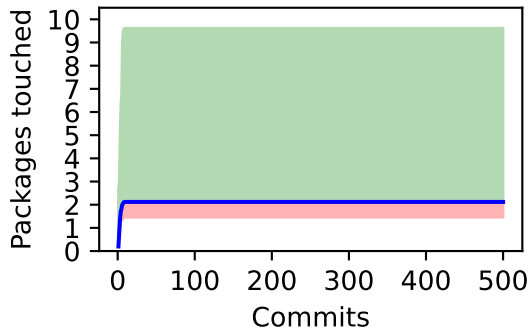


Figure 57: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

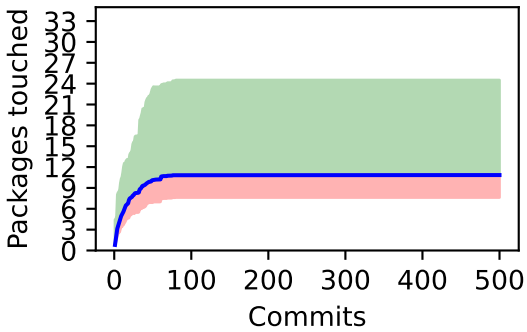
1.7.8 Time series components touched - For packages touched for each week commit



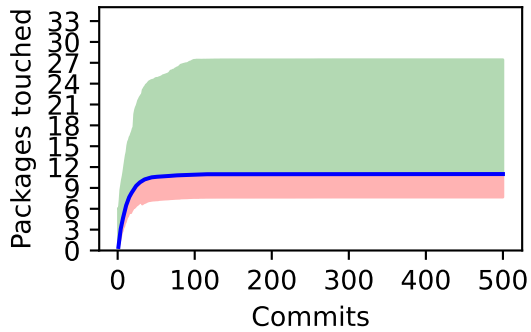
(a) Transient founder (n=28 & μ=2.93)



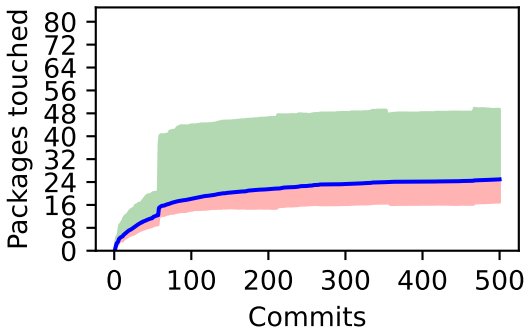
(b) Transient later joiner (n=1396 & μ=2.12)



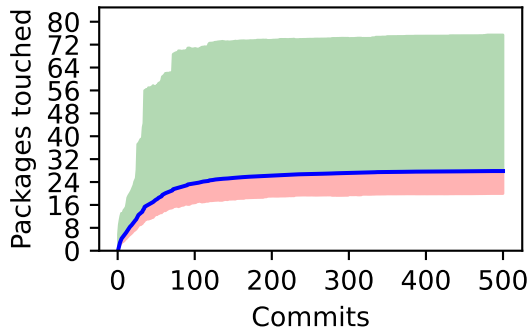
(c) Moderate founder (n=40 & μ=10.85)



(d) Moderate later joiner (n=580 & μ=10.99)

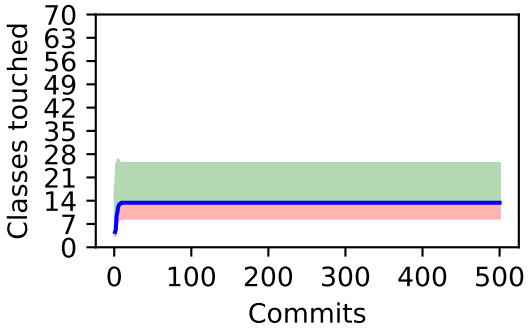


(e) Sustained founder (n=34 & μ=24.94)

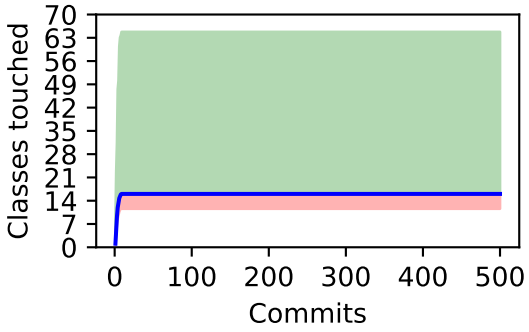


(f) Sustained later joiner (n=203 & μ=27.84)

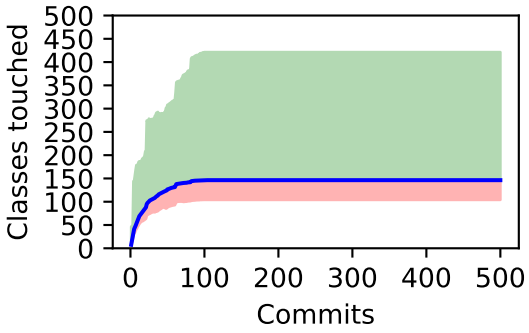
Figure 58: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



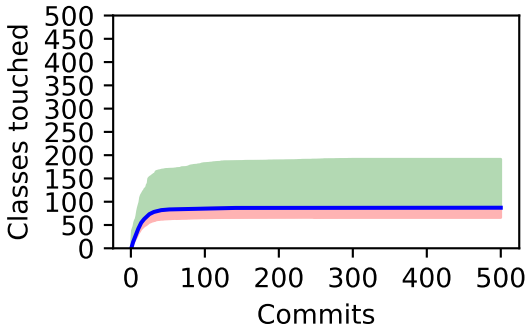
(a) Transient founder (n=28 & μ=13.39)



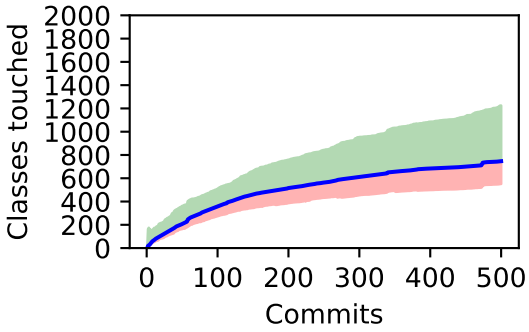
(b) Transient later joiner (n=1396 & μ=16.05)



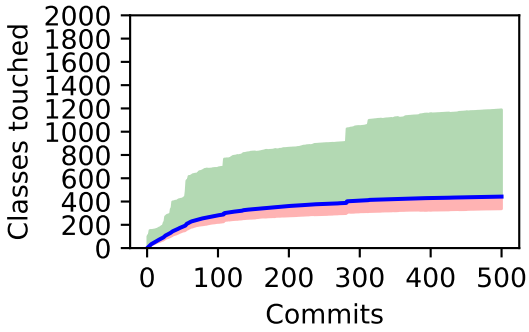
(c) Moderate founder (n=40 & μ=146.32)



(d) Moderate later joiner (n=580 & μ=87.13)

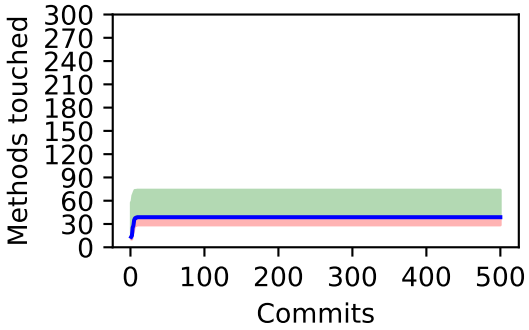


(e) Sustained founder (n=34 & μ=746.59)

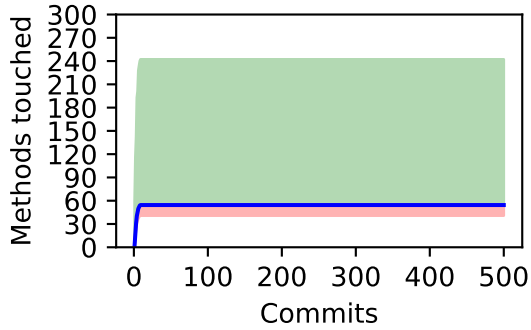


(f) Sustained later joiner (n=203 & μ=442.33)

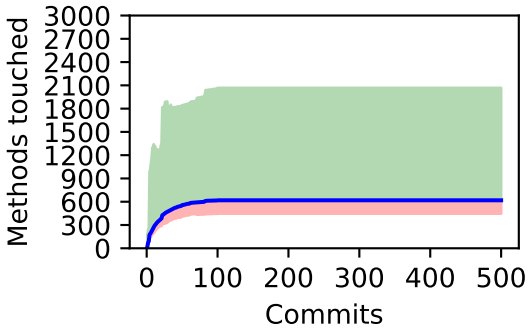
Figure 59: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



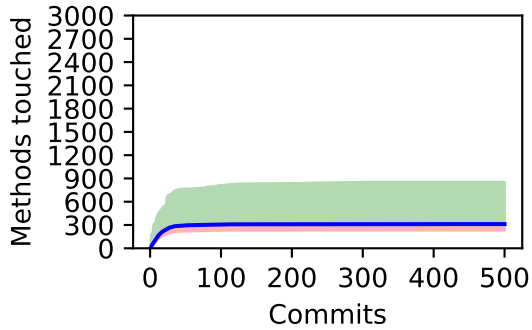
(a) Transient founder (n=28 & μ=38.75)



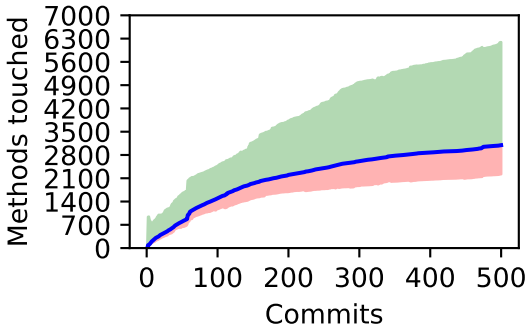
(b) Transient later joiner (n=1396 & μ=54.43)



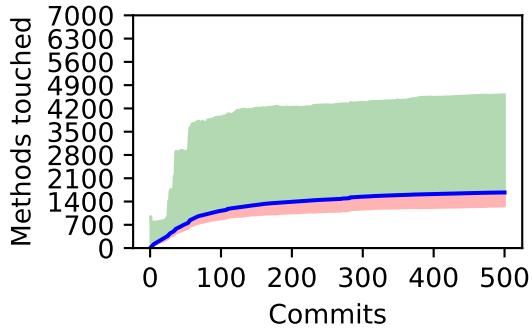
(c) Moderate founder (n=40 & μ=618.5)



(d) Moderate later joiner (n=580 & μ=311.69)



(e) Sustained founder (n=34 & μ=3095.18)



(f) Sustained later joiner (n=203 & μ=1672.84)

Figure 60: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

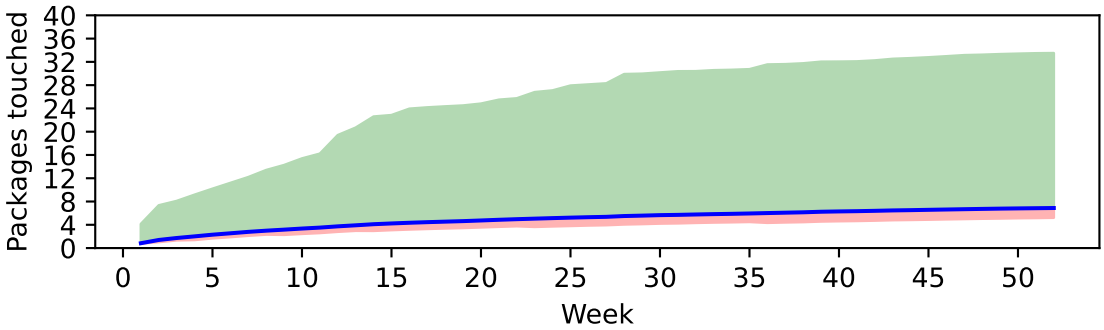


Figure 61: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for All developers (n=2278).

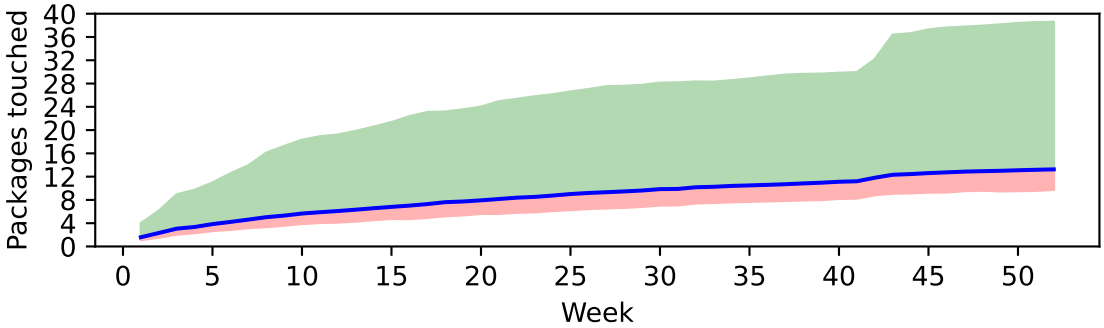


Figure 62: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for All Founders developers (n=102).

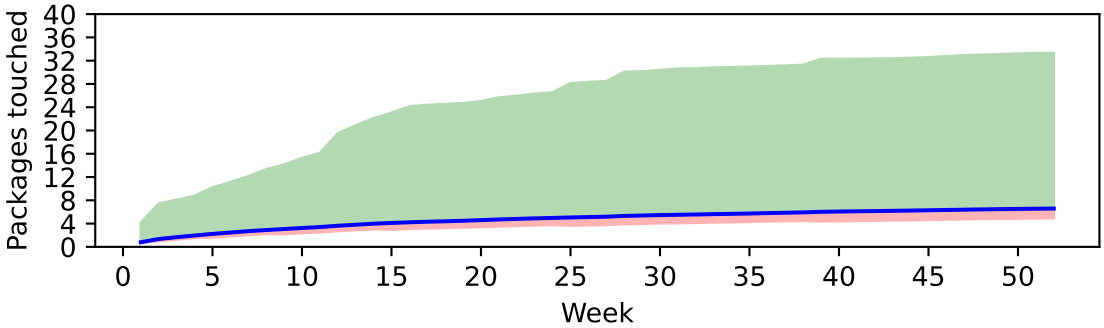


Figure 63: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for All Joiners developers (n=2177).

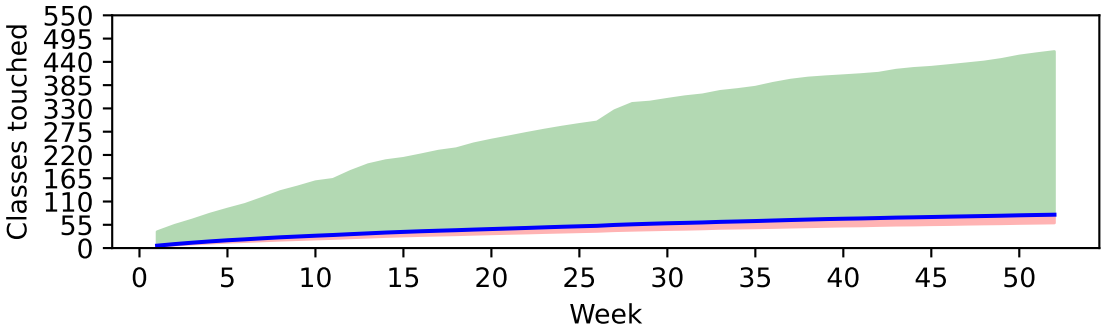


Figure 64: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for All developers (n=2278).

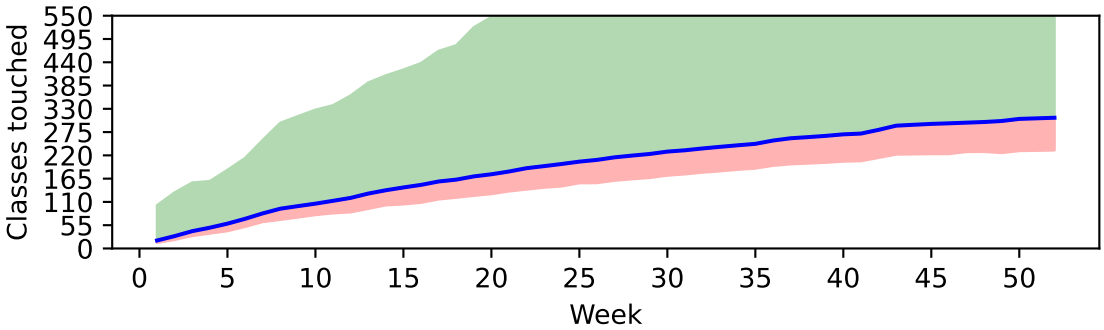


Figure 65: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for All Founders developers (n=102).

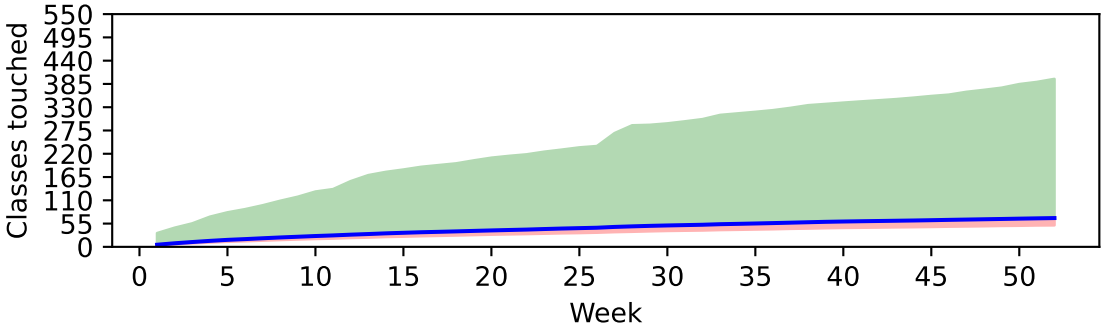


Figure 66: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for All Joiners developers (n=2177).

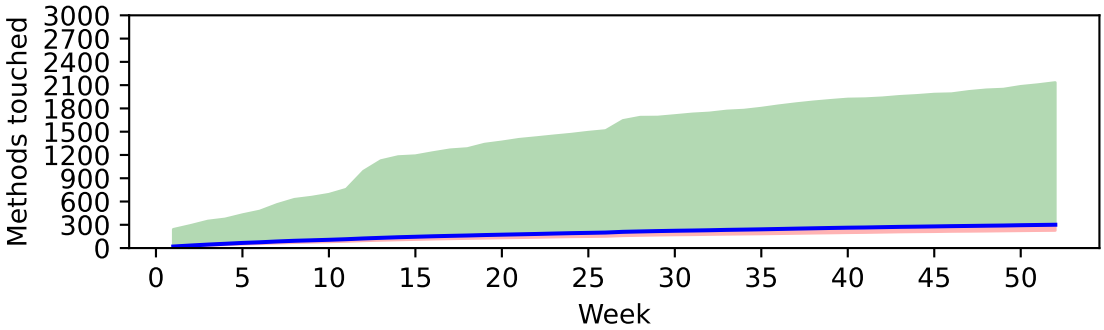


Figure 67: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for All developers (n=2278).

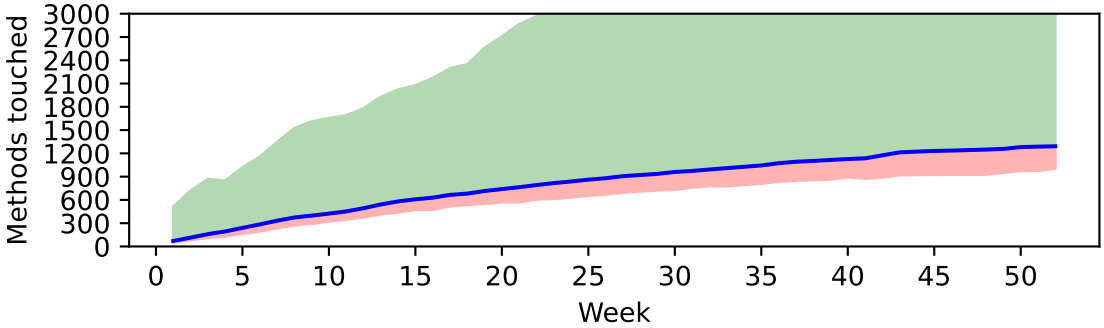


Figure 68: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for All Founders developers (n=102).

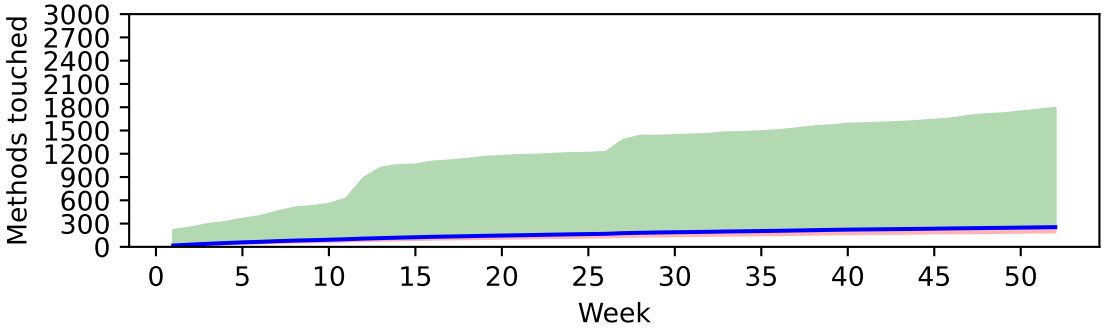
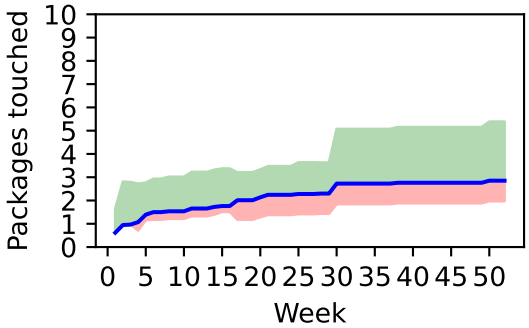


Figure 69: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for All Joiners developers (n=2177).

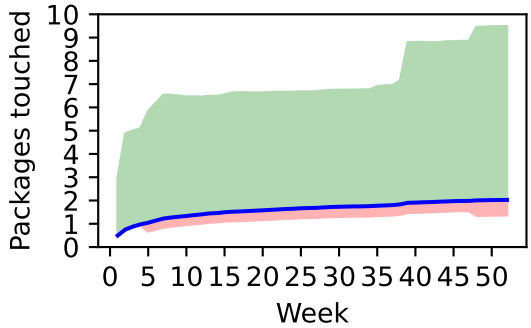
1.8 Welch t test - For all components touched for all commits

A Welch t-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances. A Welch Statistic magnitude greater than 2.0 generally indicates a significant difference ($p < 0.05$).

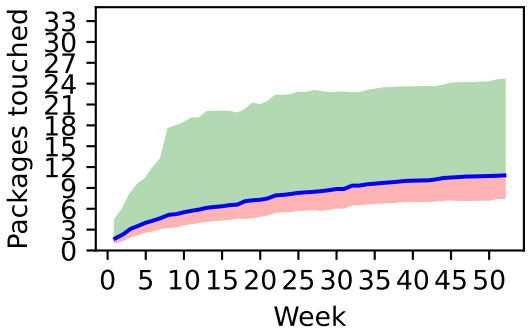
Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers. The MWU Statistic represents the number of pairwise comparisons where one group outranks the



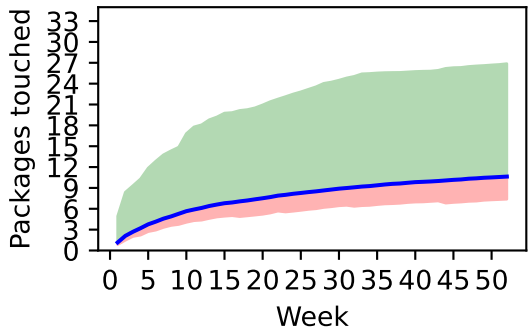
(a) Transient founder (n=28 & μ=2.85)



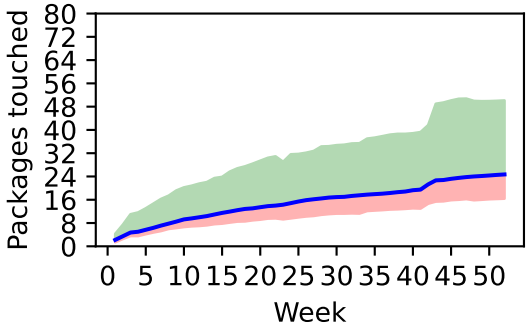
(b) Transient later joiner (n=1396 & μ=2.02)



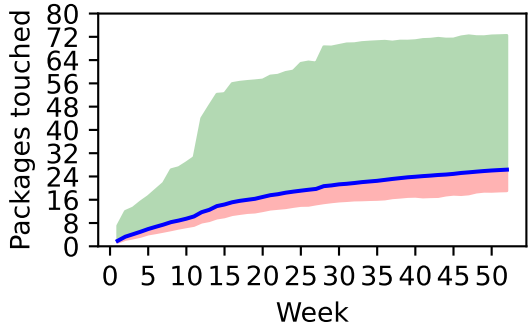
(c) Moderate founder (n=40 & μ=10.8)



(d) Moderate later joiner (n=580 & μ=10.63)

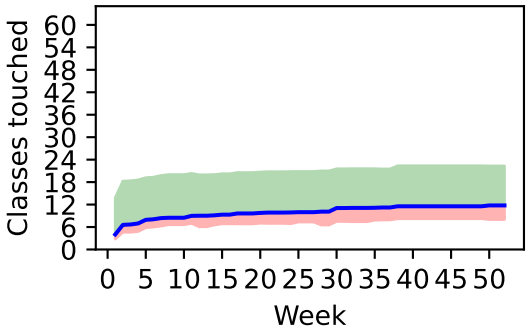


(e) Sustained founder (n=34 & μ=24.7)

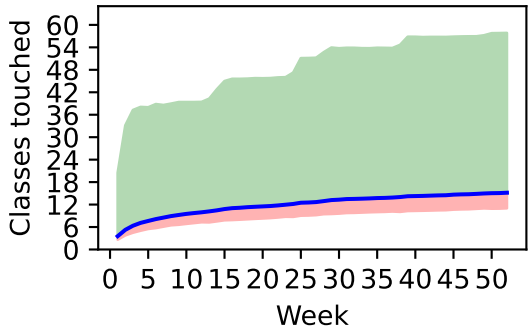


(f) Sustained later joiner (n=203 & μ=26.35)

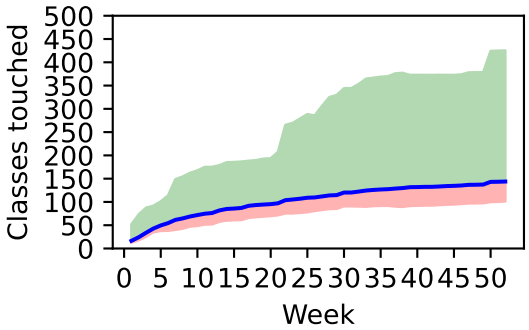
Figure 70: A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.



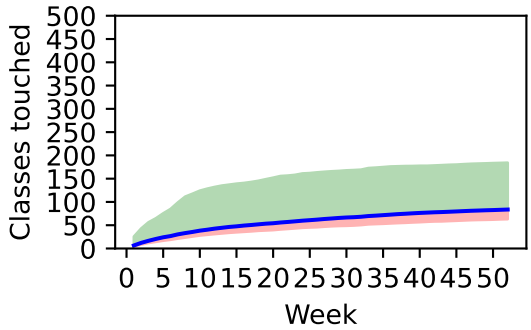
(a) Transient founder (n=28 & μ=11.77)



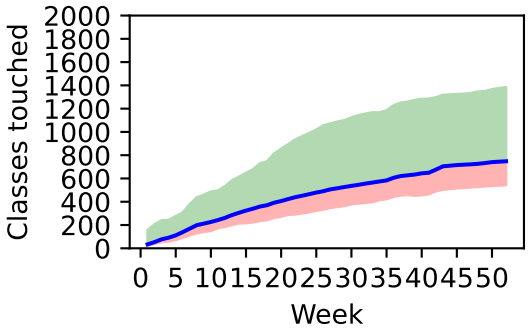
(b) Transient later joiner (n=1396 & μ=15.16)



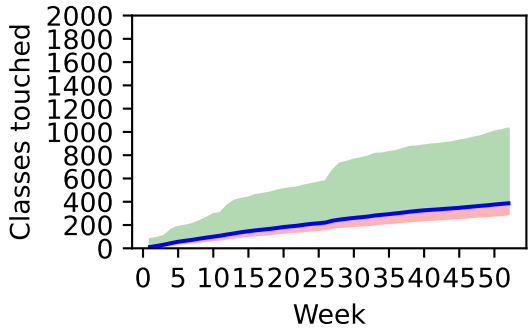
(c) Moderate founder (n=40 & μ=143.84)



(d) Moderate later joiner (n=580 & μ=83.69)

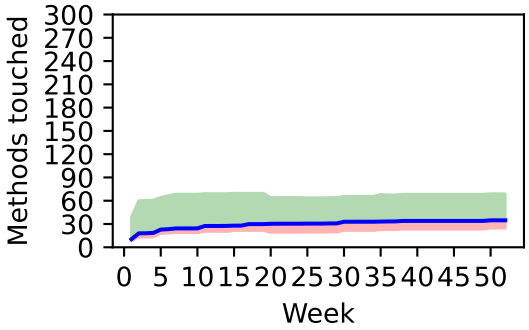


(e) Sustained founder (n=34 & μ=747.76)

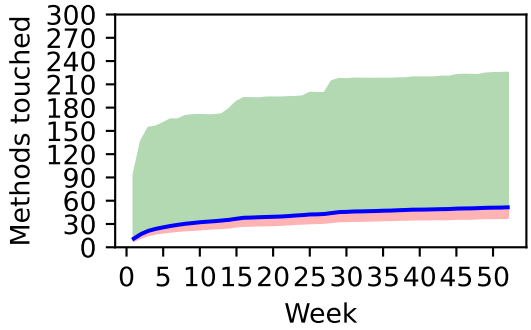


(f) Sustained later joiner (n=203 & μ=386.64)

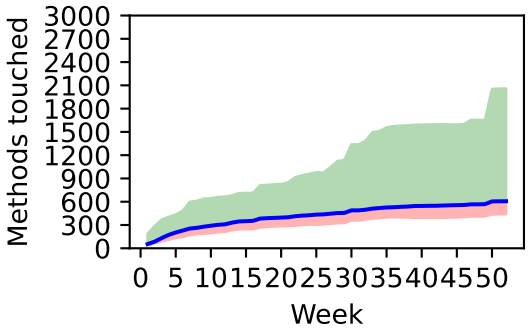
Figure 71: A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.



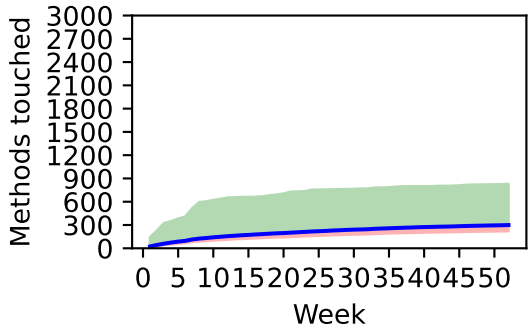
(a) Transient founder (n=28 & μ=34.74)



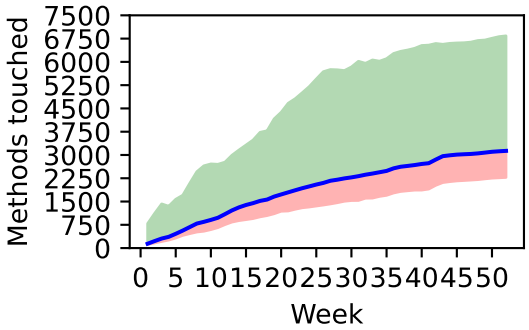
(b) Transient later joiner (n=1396 & μ=51.5)



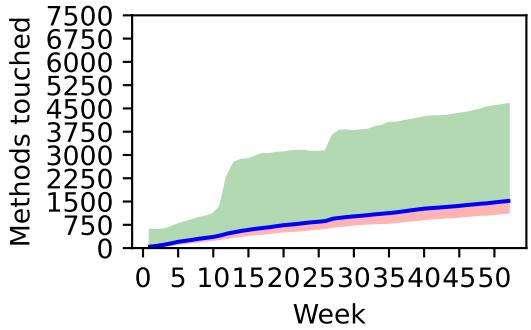
(c) Moderate founder (n=40 & μ=606.52)



(d) Moderate later joiner (n=580 & μ=299.05)



(e) Sustained founder (n=34 & μ=3132.41)



(f) Sustained later joiner (n=203 & μ=1518.36)

Figure 72: A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

other, with the total possible pairs provided in brackets. A p-value less than 0.05 is considered statistically significant.

1.8.1 Welch t test - For all components touched for all commits by Founder against Late joiner

Table 2: Welch t-test and Mann-Whitney U results for components and number of commits by Founder against Late joiner

Component	Number of First Commits	Founder μ	Late joiner μ	Welch Statistic	Welch P	MWU Statistic (Number of Pairs)	MWU P
Packages	5	3.17	2.57	1.63	0.1063	133,083.5 (222,258)	0.0004
Packages	10	4.48	3.53	1.73	0.0855	133,125.0 (222,258)	0.0004
Packages	20	6.13	4.56	2.05	0.0426	135,321.0 (222,258)	0.0001
Classes	5	31.64	17.31	2.31	0.0229	134,492.5 (222,258)	0.0003
Classes	10	50.70	25.92	3.19	0.0019	143,314.5 (222,258)	0.0000
Classes	20	73.32	35.40	4.06	0.0001	148,954.5 (222,258)	0.0000
Methods	5	129.04	60.44	1.89	0.0618	132,882.0 (222,258)	0.0008
Methods	10	201.63	90.35	2.44	0.0166	142,571.0 (222,258)	0.0000
Methods	20	293.36	122.88	3.26	0.0015	148,434.0 (222,258)	0.0000

1.8.2 Welch t test - For all components touched for all commits by Moderate against Sustained

Table 3: Welch t-test and Mann-Whitney U results for components and number of commits by Moderate against Sustained

Component	Number of First Commits	Moderate μ	Sustained μ	Welch Statistic	Welch P	MWU Statistic (Number of Pairs)	MWU P
Packages	5	3.71	2.18	6.67	0.0000	625,247.0 (1,029,820)	0.0000
Packages	10	6.03	2.66	10.28	0.0000	655,767.5 (1,029,820)	0.0000
Packages	20	8.57	3.15	12.10	0.0000	670,267.0 (1,029,820)	0.0000
Classes	5	23.29	15.95	4.92	0.0000	689,691.5 (1,029,820)	0.0000
Classes	10	43.22	20.98	10.89	0.0000	806,482.5 (1,029,820)	0.0000
Classes	20	67.03	25.92	15.18	0.0000	854,945.0 (1,029,820)	0.0000
Methods	5	83.94	55.88	3.41	0.0007	681,055.5 (1,029,820)	0.0000
Methods	10	153.35	73.67	7.39	0.0000	788,742.5 (1,029,820)	0.0000
Methods	20	237.22	90.67	11.35	0.0000	839,919.0 (1,029,820)	0.0000

1.8.3 Welch t test - For all components touched for all commits by Moderate founder against Moderate late joiner

Table 4: Welch t-test and Mann-Whitney U results for components and number of commits by Moderate founder against Moderate late joiner

Component	Number of First Commits	Moderate founder μ	Moderate late joiner μ	Welch Statistic	Welch P	MWU Statistic (Number of Pairs)	MWU P
Packages	5	3.52	2.58	1.36	0.1800	54,890.0 (89,640)	0.0101
Packages	10	5.15	3.55	1.69	0.0978	55,939.5 (89,640)	0.0045
Packages	20	7.47	4.58	2.24	0.0305	57,927.0 (89,640)	0.0008
Classes	5	41.38	17.53	2.10	0.0419	61,473.5 (89,640)	0.0001
Classes	10	61.60	26.41	2.53	0.0156	66,755.5 (89,640)	0.0000
Classes	20	86.22	36.22	3.28	0.0022	68,980.0 (89,640)	0.0000
Methods	5	186.75	61.31	1.67	0.1025	61,334.5 (89,640)	0.0001
Methods	10	271.43	92.19	1.83	0.0750	66,505.0 (89,640)	0.0000
Methods	20	379.90	126.05	2.44	0.0191	70,271.0 (89,640)	0.0000

1.8.4 Welch t test - For all components touched for all commits by Sustained founder against Sustained late joiner

Table 5: Welch t-test and Mann-Whitney U results for components and number of commits by Sustained founder against Sustained late joiner

Component	Number of First Commits	Sustained founder μ	Sustained late joiner μ	Welch Statistic	Welch P	MWU Statistic (Number of Pairs)	MWU P
Packages	5	3.18	2.58	1.07	0.2916	46,251.5 (76,398)	0.0258
Packages	10	4.97	3.55	1.38	0.1758	46,543.0 (76,398)	0.0210
Packages	20	7.18	4.59	1.72	0.0938	47,660.5 (76,398)	0.0089
Classes	5	36.32	17.67	1.51	0.1416	49,393.0 (76,398)	0.0033
Classes	10	68.59	26.40	2.80	0.0084	56,606.5 (76,398)	0.0000
Classes	20	107.50	36.03	3.93	0.0004	62,160.0 (76,398)	0.0000
Methods	5	138.53	62.37	1.23	0.2274	48,378.5 (76,398)	0.0076
Methods	10	253.65	92.93	2.33	0.0259	56,158.0 (76,398)	0.0000
Methods	20	401.24	126.41	3.13	0.0036	60,489.0 (76,398)	0.0000

1.9 Anova generate - For all components touched for all commits

An Anova table for all components touched on average.

1.9.1 Anova generate - For all components touched for a all commits including transient

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage package exploration.
- **Tenure(M/S):** Tests the *Main Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

Table 6: Anova results for compents and number of commits including transient

Component	Number of First Commits	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
packages	5	Role (F/L)	1.43	1	0.07	0.7858	0.0000
packages	5	Tenure (M/S)	12.82	2	0.33	0.7186	0.0003
packages	5	Interaction	47.70	2	1.23	0.2925	0.0011
packages	all commits including transient	Residual	44123.75	2275	-	-	
packages	10	Role (F/L)	33.04	1	0.92	0.3364	0.0004
packages	10	Tenure (M/S)	93.53	2	1.31	0.2703	0.0011
packages	10	Interaction	60.15	2	0.84	0.4311	0.0007
packages	all commits	Residual	81291.22	2275	-	-	
packages	20	Role (F/L)	51.36	1	0.88	0.3489	0.0004
packages	20	Tenure (M/S)	396.57	2	3.39	0.0339	0.0029
packages	20	Interaction	151.58	2	1.30	0.2741	0.0011
packages	all commits	Residual	133126.89	2275	-	-	
classes	5	Role (F/L)	13981.95	1	13.10	0.0003	0.0055
classes	5	Tenure (M/S)	15297.80	2	7.16	0.0008	0.0060
classes	5	Interaction	7200.09	2	3.37	0.0345	0.0028
classes	all commits	Residual	2428957.52	2275	-	-	
classes	10	Role (F/L)	14448.45	1	8.68	0.0032	0.0036
classes	10	Tenure (M/S)	54603.06	2	16.41	0.0000	0.0136
classes	10	Interaction	10049.57	2	3.02	0.0490	0.0025
classes	all commits	Residual	3785966.86	2275	-	-	
classes	20	Role (F/L)	15761.21	1	6.84	0.0090	0.0028
classes	20	Tenure (M/S)	146938.17	2	31.87	0.0000	0.0257
classes	20	Interaction	12952.28	2	2.81	0.0605	0.0023
classes	all commits	Residual	5244492.84	2275	-	-	
methods	5	Role (F/L)	451979.70	1	17.19	0.0000	0.0073
methods	5	Tenure (M/S)	383522.02	2	7.29	0.0007	0.0062
methods	5	Interaction	254925.91	2	4.85	0.0079	0.0041
methods	all commits	Residual	59824573.45	2275	-	-	
methods	10	Role (F/L)	596144.39	1	15.72	0.0001	0.0065
methods	10	Tenure (M/S)	1029687.05	2	13.58	0.0000	0.0113
methods	10	Interaction	319730.20	2	4.22	0.0149	0.0035
methods	all commits	Residual	86259457.30	2275	-	-	
methods	20	Role (F/L)	870509.38	1	17.91	0.0000	0.0072
methods	20	Tenure (M/S)	2510362.61	2	25.83	0.0000	0.0209
methods	20	Interaction	460313.59	2	4.74	0.0089	0.0038
methods	all commits	Residual	110569092.31	2275	-	-	

1.9.2 Anova generate - For all components touched for a all commits excluding tranient

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage package exploration.
- **Tenure(M/S):** Tests the *Main Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

Table 7: Anova results for compents and number of commits excluding transient

Component	Number of First Commits	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
packages	5	Role (F/L)	1.43	1	0.05	0.8250	0.0001
packages	5	Tenure (M/S)	2.23	1	0.08	0.7826	0.0001
packages	5	Interaction	11.68	1	0.40	0.5279	0.0005
packages	all commits excluding tranient	Residual	24984.13	853	-	-	
packages	10	Role (F/L)	33.04	1	0.59	0.4443	0.0007
packages	10	Tenure (M/S)	0.59	1	0.01	0.9185	0.0000
packages	10	Interaction	0.01	1	0.00	0.9889	0.0000
packages	all commits	Residual	48127.58	853	-	-	
packages	20	Role (F/L)	51.36	1	0.44	0.5081	0.0005
packages	20	Tenure (M/S)	1.64	1	0.01	0.9059	0.0000
packages	20	Interaction	26.53	1	0.23	0.6344	0.0003
packages	all commits	Residual	99963.25	853	-	-	
classes	5	Role (F/L)	13981.95	1	7.35	0.0069	0.0082
classes	5	Tenure (M/S)	468.97	1	0.25	0.6198	0.0003
classes	5	Interaction	2774.71	1	1.46	0.2276	0.0016
classes	all commits	Residual	1623543.63	853	-	-	
classes	10	Role (F/L)	14448.45	1	5.02	0.0254	0.0055
classes	10	Tenure (M/S)	897.52	1	0.31	0.5769	0.0003
classes	10	Interaction	15.03	1	0.01	0.9424	0.0000
classes	all commits	Residual	2457107.31	853	-	-	
classes	20	Role (F/L)	15761.21	1	3.43	0.0642	0.0037
classes	20	Tenure (M/S)	8318.53	1	1.81	0.1786	0.0020
classes	20	Interaction	420.20	1	0.09	0.7623	0.0001
classes	all commits	Residual	3915633.28	853	-	-	
methods	5	Role (F/L)	451979.70	1	8.27	0.0041	0.0093
methods	5	Tenure (M/S)	42733.87	1	0.78	0.3769	0.0009
methods	5	Interaction	121524.53	1	2.22	0.1364	0.0025
methods	all commits	Residual	46646591.43	853	-	-	
methods	10	Role (F/L)	596144.39	1	7.60	0.0060	0.0085
methods	10	Tenure (M/S)	5808.58	1	0.07	0.7856	0.0001
methods	10	Interaction	46476.28	1	0.59	0.4417	0.0007
methods	all commits	Residual	66927848.79	853	-	-	
methods	20	Role (F/L)	870509.38	1	8.14	0.0044	0.0089
methods	20	Tenure (M/S)	8365.74	1	0.08	0.7798	0.0001
methods	20	Interaction	35239.39	1	0.33	0.5661	0.0004
methods	all commits	Residual	91237483.81	853	-	-	

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{1}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{2}$$

1.10 Strategy logistic regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^{β}):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **50% Threshold:** The threshold number of components to touch to have a 50
- **Moderate Met Threshold:** The moderate number of developers to have met the threshold, the percentate of developers in brackets.
- **Sustained Met Threshold:** The sustained number of developers to have met the threshold, the percentate of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 8: Strategy Logistic Regression for all components against number of commits

Component	Number of First Commits	Type	Moderate μ	Sustained μ	Coefficient	Odds Ratio	Std. Err	p-Value	30% Threshold	Moderate Met Threshold	Sustained Met Threshold
packages	5	Intercept	3.30	16.48	-2.6487	0.0707	0.0878	0.0000	-	-	-
packages	5	Packages Count	3.30	16.48	0.0711	1.0737	0.0059	0.0000	**25.3339**	43 (2.1%)	53 (22.4%)
packages	10	Intercept	4.23	23.71	-2.7638	0.0631	0.0921	0.0000	-	-	-
packages	10	Packages Count	4.23	23.71	0.0633	1.0653	0.0048	0.0000	**30.2976**	47 (2.3%)	65 (27.4%)
packages	20	Intercept	4.47	25.81	-2.7951	0.0611	0.0931	0.0000	-	-	-
packages	20	Packages Count	4.47	25.81	0.0613	1.0632	0.0045	0.0000	**31.7932**	53 (2.6%)	71 (30.0%)
classes	5	Intercept	24.57	214.75	-3.2138	0.0402	0.1103	0.0000	-	-	-
classes	5	Classes Count	24.57	214.75	0.0162	1.0163	0.0010	0.0000	**146.3910**	54 (2.6%)	101 (42.6%)
classes	10	Intercept	33.49	381.41	-3.8441	0.0214	0.1420	0.0000	-	-	-
classes	10	Classes Count	33.49	381.41	0.0161	1.0162	0.0009	0.0000	**186.5769**	53 (2.6%)	146 (61.6%)
classes	20	Intercept	35.63	427.28	-3.9773	0.0187	0.1500	0.0000	-	-	-
classes	20	Classes Count	35.63	427.28	0.0156	1.0157	0.0009	0.0000	**200.3186**	51 (2.5%)	151 (63.7%)
methods	5	Intercept	88.19	878.82	-2.9725	0.0512	0.0996	0.0000	-	-	-
methods	5	Methods Count	88.19	878.82	0.0034	1.0034	0.0002	0.0000	**629.3813**	47 (2.3%)	90 (38.0%)
methods	10	Intercept	119.63	1515.71	-3.5013	0.0302	0.1244	0.0000	-	-	-
methods	10	Methods Count	119.63	1515.71	0.0035	1.0035	0.0002	0.0000	**751.7739**	46 (2.3%)	131 (55.3%)
methods	20	Intercept	127.59	1704.54	-3.6114	0.0270	0.1301	0.0000	-	-	-
methods	20	Methods Count	127.59	1704.54	0.0034	1.0034	0.0002	0.0000	**808.8911**	47 (2.3%)	139 (58.6%)

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon \tag{3}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{4}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy.

2 Repository: 14

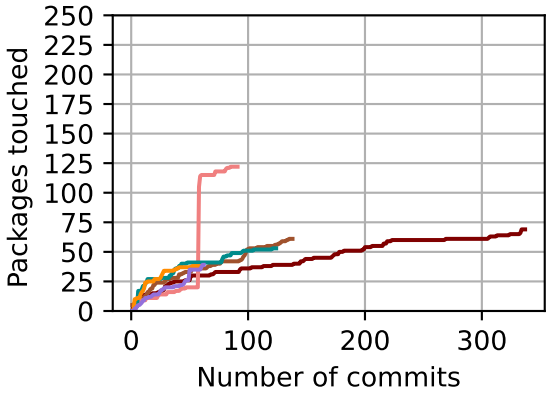
2.1 Time series sustained components touched by developer - 14 For developer commits and components touched.

Table 9: Table of first 6 sustained founder developers in the Scatter Developer graphs.

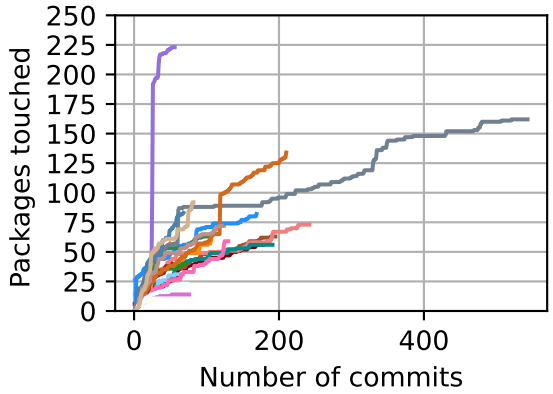
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

Table 10: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

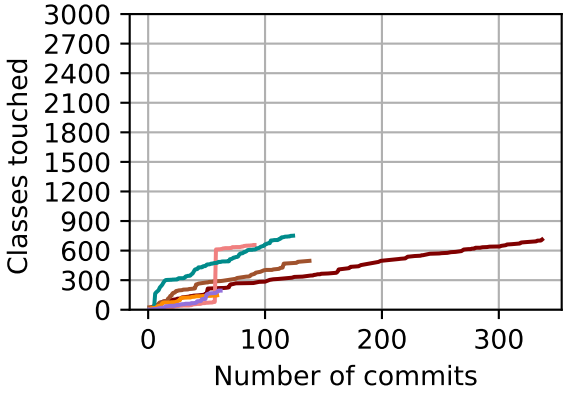
Developer ID	Colour
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



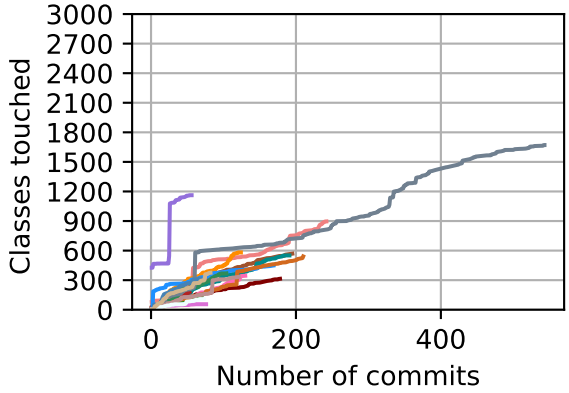
(a) Packages for sustained founder



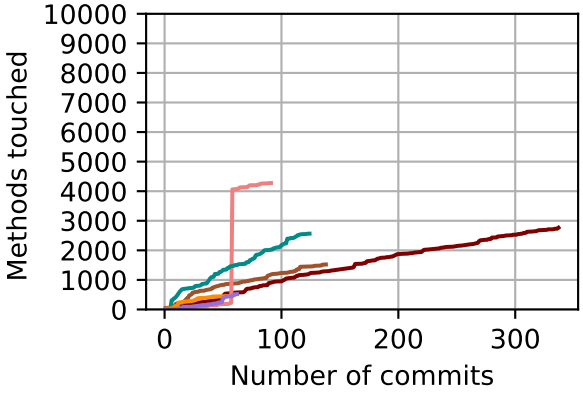
(b) Packages for sustained later joiner



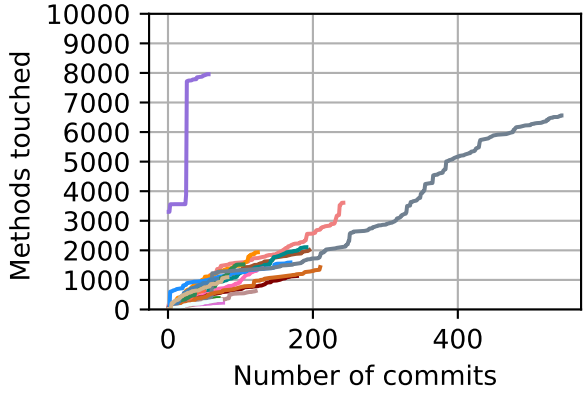
(c) Classes for sustained founder



(d) Classes for sustained later joiner



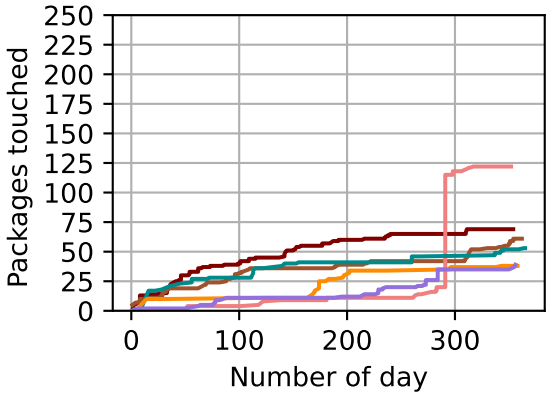
(e) Methods for sustained founder



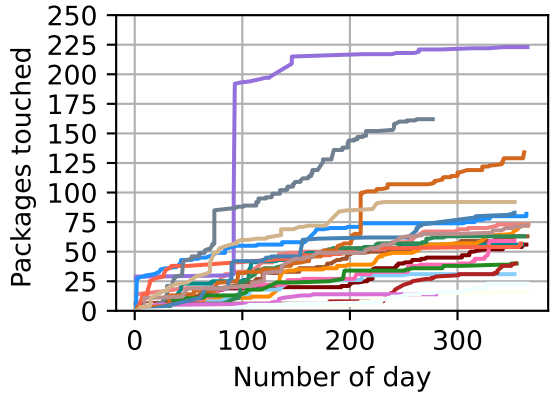
(f) Methods for sustained later joiner

Figure 73: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 9. Sustained later joiner developers with colours are shown in Table 10.

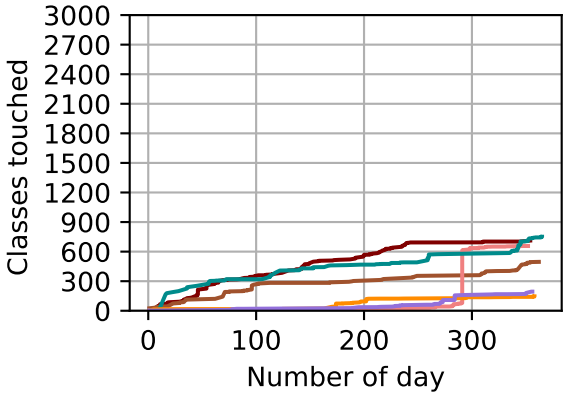
2.2 Time series sustained components touched by developer - 14 For developer day and components touched.



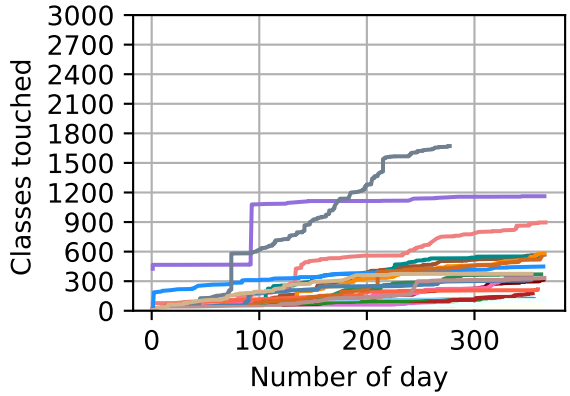
(a) Packages for sustained founder



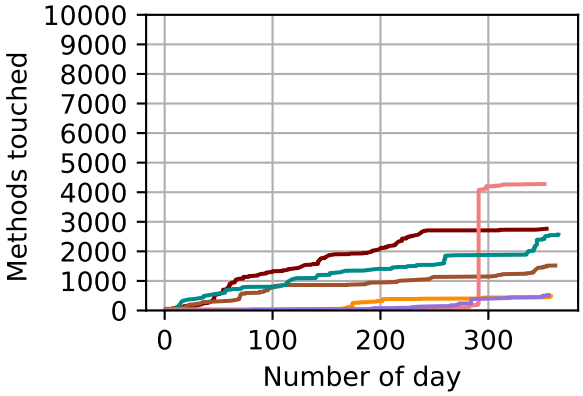
(b) Packages for sustained later joiner



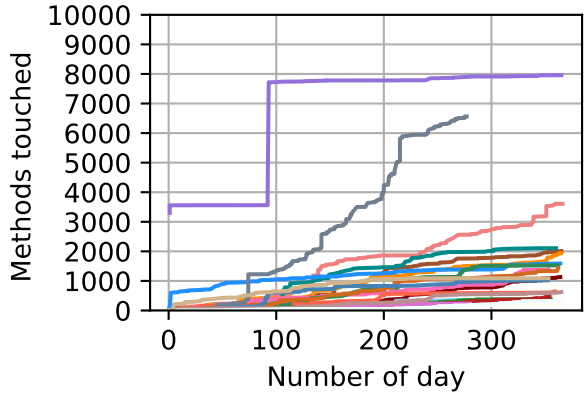
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 74: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 9. Sustained later joiner developers with colours are shown in Table 10.

3 Repository: 21

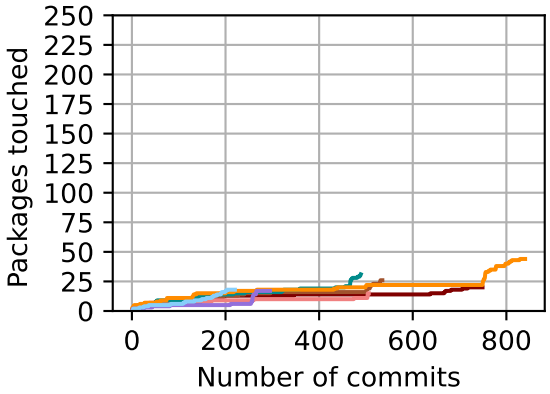
3.1 Time series sustained components touched by developer - 21 For developer commits and components touched.

Table 11: Table of first 7 sustained founder developers in the Scatter Developer graphs.

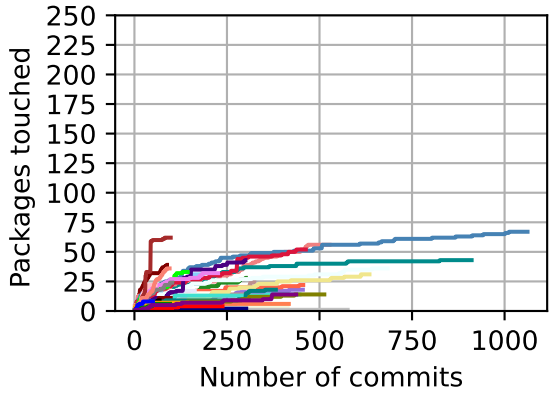
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 12: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

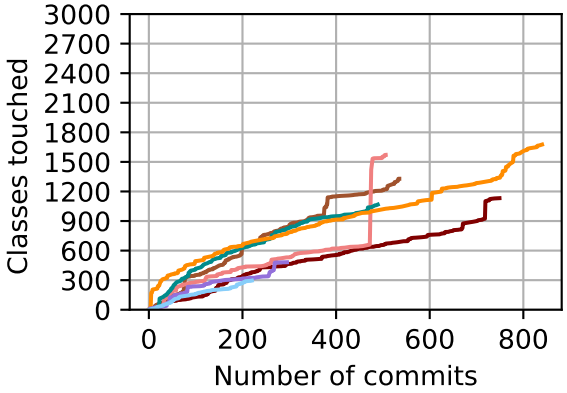
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



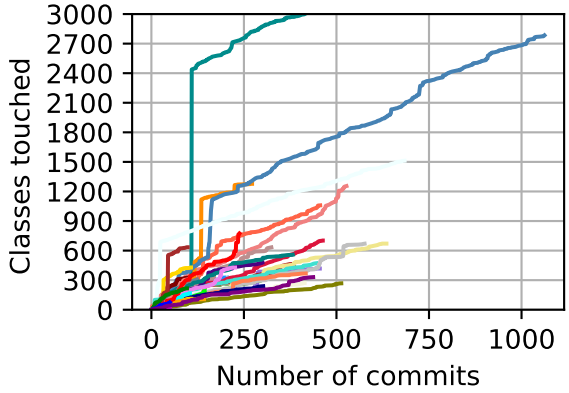
(a) Packages for sustained founder



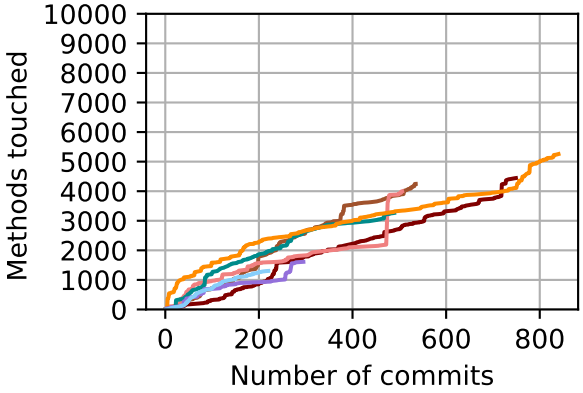
(b) Packages for sustained later joiner



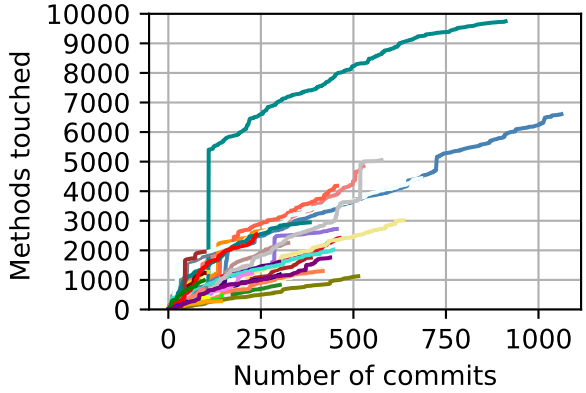
(c) Classes for sustained founder



(d) Classes for sustained later joiner



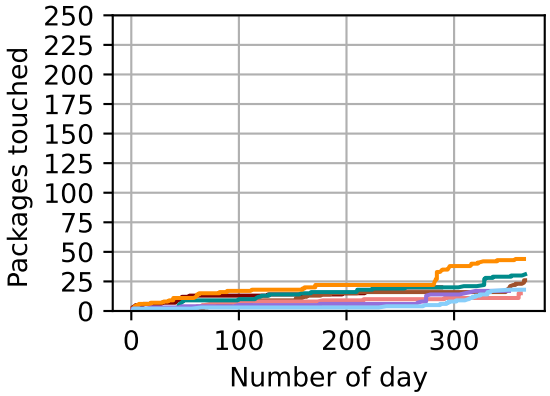
(e) Methods for sustained founder



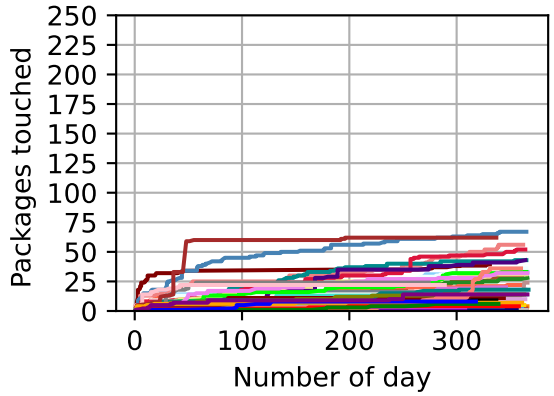
(f) Methods for sustained later joiner

Figure 75: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 11. Sustained later joiner developers with colours are shown in Table 12.

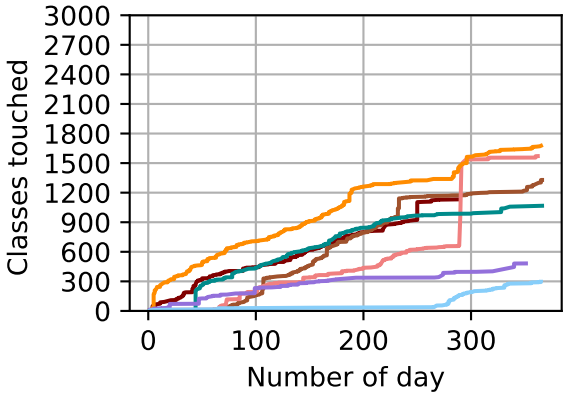
3.2 Time series sustained components touched by developer - 21 For developer day and components touched.



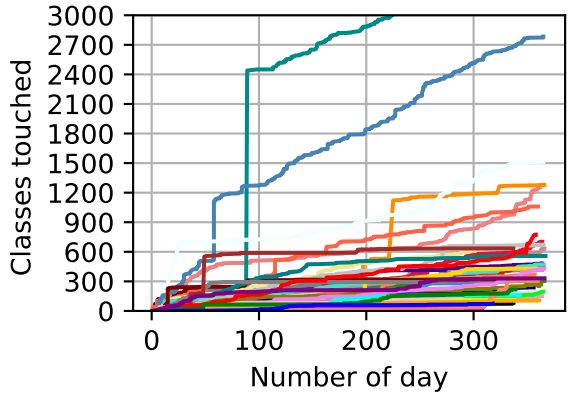
(a) Packages for sustained founder



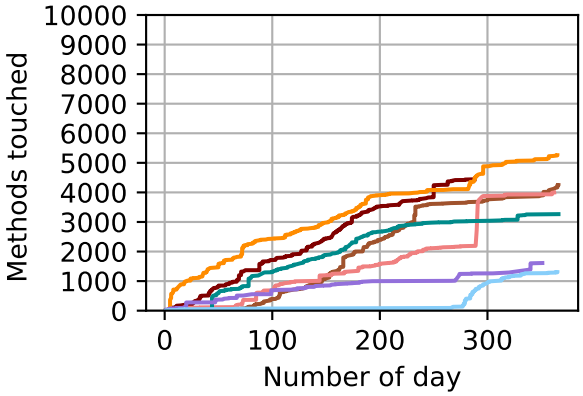
(b) Packages for sustained later joiner



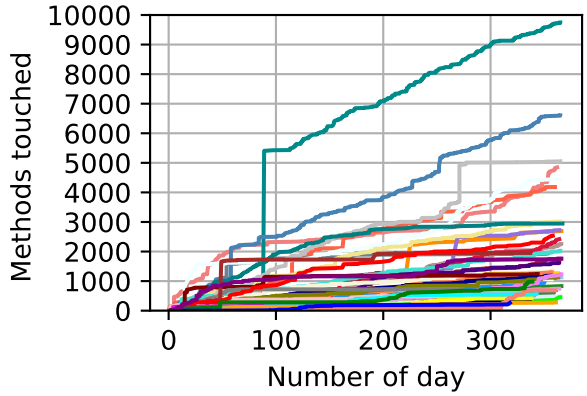
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 76: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 11. Sustained later joiner developers with colours are shown in Table 12.